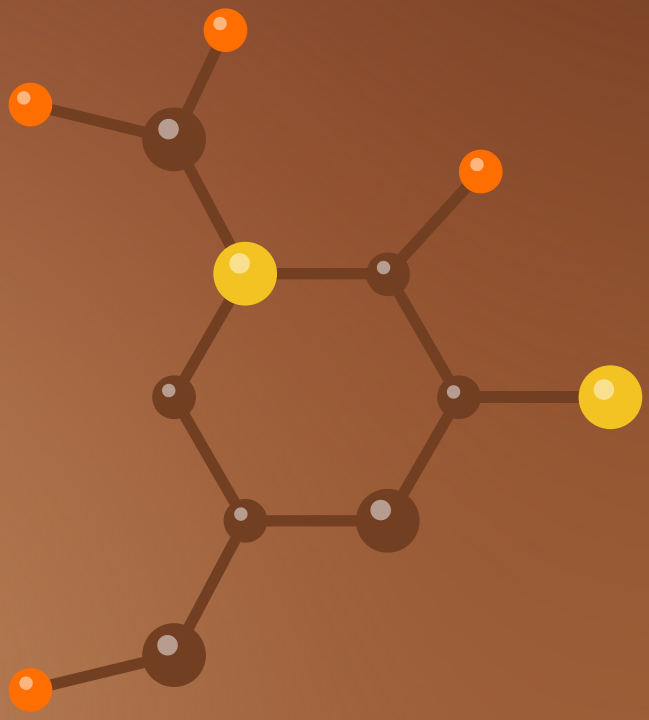
ALLEN CAREER INSTITUTE
path to success
KOTA (RAJASTHAN)



ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Aromatic Compounds

English Medium

AROMATIC COMPOUNDS

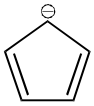
EXERCISE # O-I

1. Which of the following group is divalent:

(A) Benzoyl (B) Benzyl (C) Benzal (D) p-Tolyl

AH0002

2. How many π electron are there in the following species :



(A) 2 (B) 4 (C) 6 (D) 8

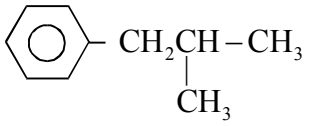
AH0004

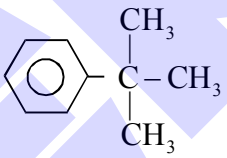
3. The number of benzylic hydrogen atoms in ethylbenzene is:

(A) 3 (B) 5 (C) 2 (D) 7

AH0005

4.  + $CH_3CH_2CH_2CH_2Cl \xrightarrow{AlCl_3}$ hydrocarbon (X) major product X is:

(A) 

(B) 

(C) 

(D) None is correct

AH0007

5. In the sulphonation, acetylation and formylation of benzene the group of effective electrophiles would be :

(A) $SO_3^+, CH_3C \equiv \overset{+}{O}H, \overset{+}{C}HO$ (B) $SO_3, CH_3 - C \equiv \overset{+}{O}H, \overset{+}{C}HO$

(C) SO_3, CH_3CHO, CO + HCl (D) HSO_3, CH_3CO, HCO

AH0008

6. o/p ratio in highest for nitration of which of the following compound ?

(A) Ethyl benzene (B) Toluene

(C) Isopropyl benzene (D) Tertiarybutyl benzene

AH0009

7. Which can be used to generate NO_2^+ in nitration of benzene ring

(A) $HNO_3 + H_2SO_4$ (B) $HNO_3 + HClO_4$ (C) N_2O_5 (D) All

AH0010

8. For the electrophilic substitution reaction involving nitration, which of the following sequence regarding the rate of reaction is true?

(A) $k_{C_6H_6} > k_{C_6D_6} > k_{C_6T_6}$ (B) $k_{C_6H_6} < k_{C_6D_6} < k_{C_6T_6}$

(C) $k_{C_6H_6} = k_{C_6D_6} = k_{C_6T_6}$ (D) $k_{C_6H_6} > k_{C_6D_6} < k_{C_6T_6}$

AH0011

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9. For the electrophilic substitution reaction involving sulphonation, which of the following sequence regarding the rate of reaction is true?

(A) $k_{C_6H_6} > k_{C_6D_6} > k_{C_6T_6}$ (B) $k_{C_6H_6} < k_{C_6D_6} < k_{C_6T_6}$

(C) $k_{C_6H_6} = k_{C_6D_6} = k_{C_6T_6}$ (D) $k_{C_6H_6} > k_{C_6D_6} < k_{C_6T_6}$

AH0012

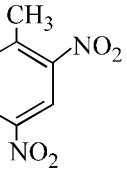
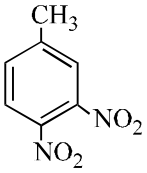
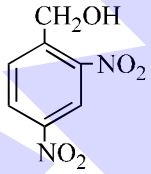
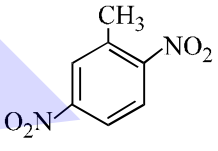
10. $C_6H_6 \xrightarrow[AlCl_3]{CH_3COCl} A \xrightarrow[HCl]{Zn-Hg} B$

The end product in the above sequence is:

(A) Toluene (B) Ethyl benzene (C) Both the above (D) None

AH0013

11. p-Nitrotoluene on further nitration gives:

(A)  (B)  (C)  (D) 

AH0014

12. Reaction of SO_3 is easier in:

(A) Benzene (B) Toluene (C) Nitrobenzene (D) chlorobenzene

AH0015

13. Which order is correct for the decreasing reactivity to ring monobromination of the following compounds:

(I) $C_6H_5CH_3$ (II) C_6H_5COOH (III) C_6H_6 (IV) $C_6H_5NO_2$

(A) I > II > III > IV (B) I > III > II > IV (C) II > III > IV > I (D) III > I > II > IV

AH0016

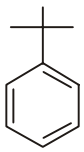
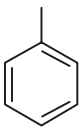
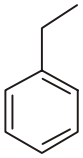
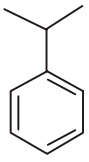
14. The highest yield of m-product is possible by the electrophilic substitution of the following:

(A) $C_6H_5CH_3$ (B) $C_6H_5CH_2COOC_2H_5$

(C) $C_6H_5CH(COOC_2H_5)_2$ (D) $C_6H_5C(COOC_2H_5)_3$

AH0017

15. Which of the following will undergo sulphonation at fastest rate ?

(A)  (B)  (C)  (D) 

AH0018

16. Aniline under acidic medium, when chlorinated, produces:

(A) o-Chloro aniline (B) m-Chloro aniline

(C) p-Chloro aniline (D) Mixture of ortho and para-chloro aniline

AH0019

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17. When sulphonilic acid (p-$H_2NC_6H_4SO_3H$) is treated with excess of bromine water, the product is:
(A) Tribromo product (B) Dibromo product
(C) Monobromo product (D) Tetrebromo product

AH0020

18. In a reaction of C_6H_5Y, the major product (>60%) is m-isomer, so the group Y is:
(A) –COOH (B) –Cl (C) –OH (D) $–NH_2$

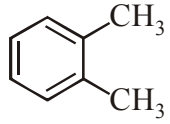
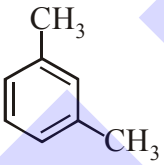
AH0021

19. An aromatic compound of molecular formula $C_6H_4Br_2$ was nitrated then three isomers of formula $C_6H_3Br_2NO_2$ were obtained. The original compound is:
(A) o-Dibromobenzene (B) m-Dibromobenzene
(C) p-Dibromobenzene (D) Both A & C

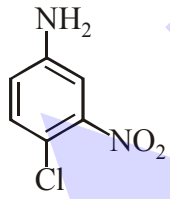
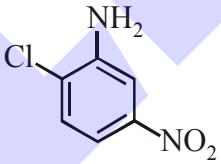
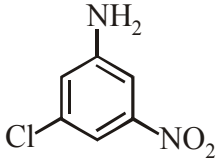
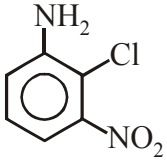
AH0022

20. Which of the following is most reactive towards sulphonation?
(A) m-Xylene (B) o-Xylene (C) Toluene (D) p-Xylene

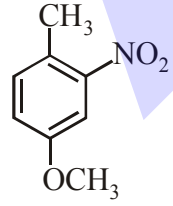
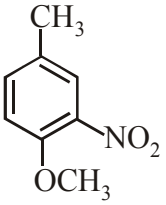
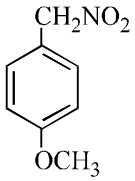
AH0023

21. Ring nitration of dimethyl benzene results in the formation of only one nitro dimethyl benzene. The dimethyl benzene is:
(A)  (B)  (C)  (D) None of these

AH0024

22. If meta-nitroaniline is chlorinated, the major product is:
(A)  (B)  (C)  (D) 

AH0025

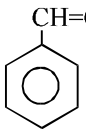
23. If p-methoxy toluene is nitrated, the major product is:
(A)  (B)  (C)  (D) No reaction

AH0026

24. $C_6H_5CH_3 \xrightarrow{CrO_2Cl_2} A \xrightarrow{H_2O} B$
The functional group present in B and name of the reaction would be
(A) –CHO, Gattermann aldehyde synthesis (B) –CHO, Etard reaction
(C) $–COCH_3$, Friedel Crafts reaction (D) –CHO, Oxo reaction

AH0027

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25. A ← Br_2  $\xrightarrow{KMnO_4}$ B

Compound A and B respectively are:

(A) o-Bromostyrene, benzoic acid

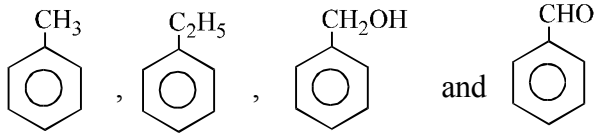
(B) p-Bromostyrene, benzaldehyde

(C) m-Bromostyrene, benzaldehyde

(D) Styrene dibromide, benzoic acid

AH0028

26. If the mixture of the following four aromatic compounds on oxidation by strong oxidising agent gives:



(A) Mixture of $C_6H_5CH_2OH + C_6H_5COOH$

(B) Mixture of $C_6H_5CHO + C_6H_5COOH$

(C) Only C_6H_5COOH

(D) None of the above

AH0029

27. Which of the following is/are produced when a mixture of benzene vapour and oxygen is passed over V_2O_5 catalyst at 775 K?

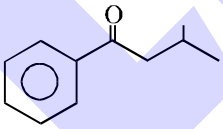
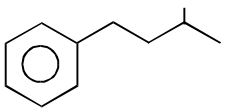
(A) Oxalic acid

(B) Glyoxal

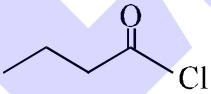
(C) Fumaric acid

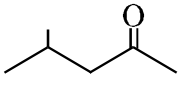
(D) Maleic anhydride

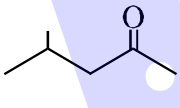
AH0030

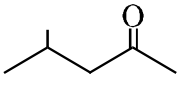
28. Benzene on reaction with 'A' forms  which on reaction with 'B' forms 

'A' and 'B' are:

(A) Zn(Hg) + conc. HCl, 

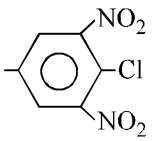
(B) 

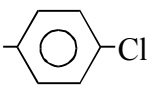
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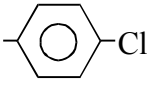
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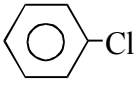
AH0031

29. Which chloroderivative of benzene among the following would undergo-hydrolysis most readily with aNaOH to furnish the corresponding hydroxy derivative.

(A) 

(B) 

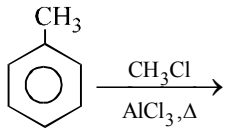
(C) 

(D) 

AH0032

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30. Major product of this reaction will be :



(A) o-Xylene (B) p-Xylene (C) Both (D) m-Xylene

AH0033

31. For preparing monoalkyl benzene, acylation process is preferred than direct alkylation because

(A) In alkylation, a poisonous gas is evolved

(B) In alkylation, large amount of heat is evolved

(C) In alkylation, polyalkylated product is formed

(D) Alkylation is very costly

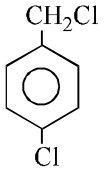
AH0034

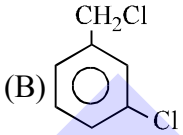
32. Phenol and ethanol are distinguished by the reaction with

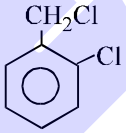
(A) Red litmus (B) $NaHCO_3$ (C) $FeCl_3$ (D) Na

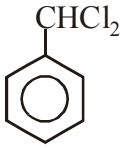
AH0035

33. An aromatic compound 'A' $C_7H_6Cl_2$, gives AgCl on bonding with alcoholic $AgNO_3$ solution, and yields C_7H_7OCl on treatment with sodium hydroxide. 'A' on oxidation gives a mono chlorobenzoic acid which affords only one mononitro derivative. The compound A is:

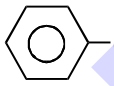
(A) 

(B) 

(C) 

(D) 

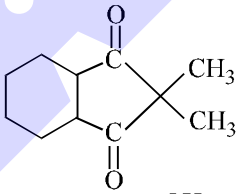
AH0036

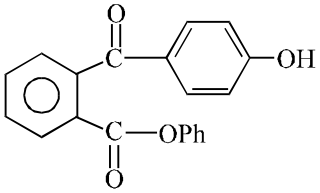
34. Chloral +  $\xrightarrow{\text{Conc. } H_2SO_4}$ product. The product is:

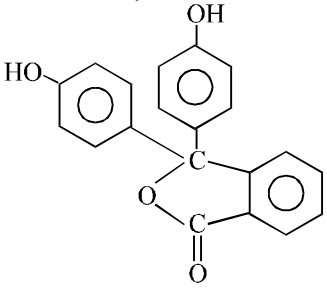
(A) Lindane (B) DDT (C) Teflon (D) Ethaneperchlorate

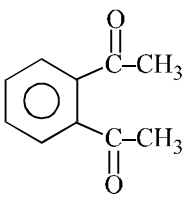
AH0038

35. Acetophenone $\xrightarrow{HCO_3H}$ A $\xrightarrow{H_3O^+}$ B + C $\xrightarrow[H^+]{\text{Pthalic Anhydride}}$ Indicator (D) ; C & D are

(A) CH_3OH & 

(B) PhOH & 

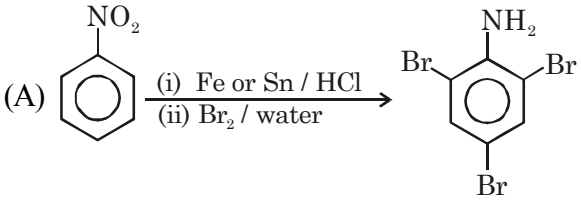
(C) PhOH & 

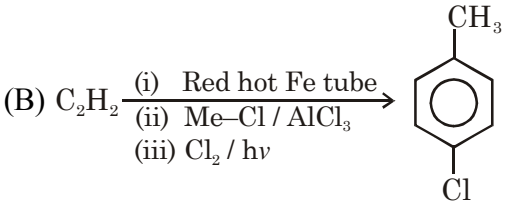
(D) CH_3OH & 

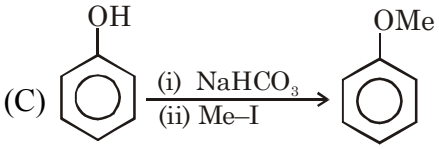
AH0039

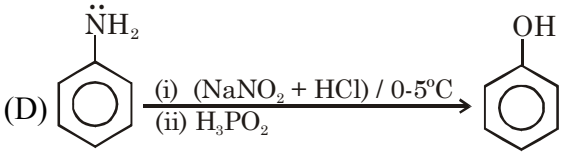
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic Aromatic Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex..p65

36. Select the reaction giving correct major product :

(A) 

(B) 

(C) 

(D) 

AH0040

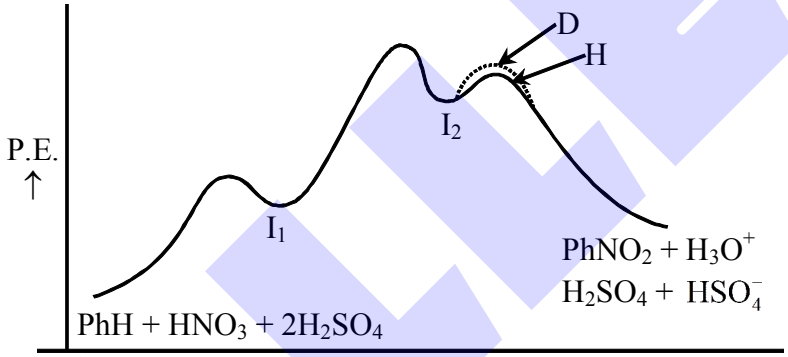
37. Phenol $\xrightarrow[(ii)\,CO_2/140°C]{(i)\,NaOH}$ A $\xrightarrow{H^+/H_2O}$ B $\xrightarrow[CH_3COOH,\Delta]{Al_2O_3}$ C

In this reaction, the end product C is:

(A) Salicylaldehyde (B) Salicylic acid (C) Phenyl acetate (D) Aspirin

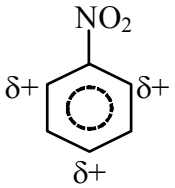
AH0041

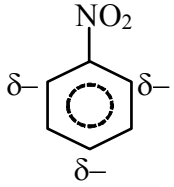
38. During nitration of benzene ring using acid mixture the following energy diagram is used.

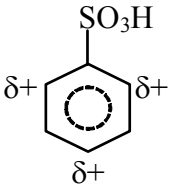


I_2 represents:

(A) $NO_2^+ + HSO_4^-$

(B) σ complex 

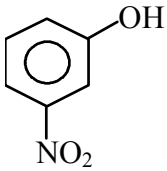
(C) 

(D) 

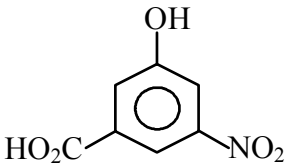
AH0227

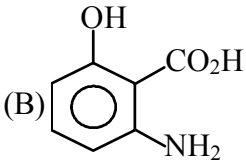
ALLEN®

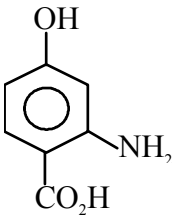
noida06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

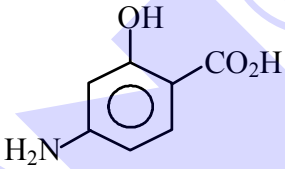
39.  $\xrightarrow{Sn+HCl}$ $\xrightarrow[+CO_2/H^+]{NaOH}$ final product (major) P

Product P is:

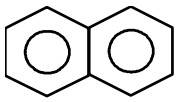
(A) 

(B) 

(C) 

(D) 

AH0228

40.  $\xrightarrow[160°C]{H_2SO_4}$ $\xrightarrow[ii)H_2O]{(i)NaOH(fused)}$ $\xrightarrow[H_2O]{CHCl_3+NaOH}$ Q (major).

Q is also formed by which of the following reaction.

(A) α-napthol when undergoes reaction with $CHCl_3$ + NaOH/aq.

(B) α-napthol when reacts with HCN + $HCl/ZnCl_2$ followed by hydrolysis

(C) β-napthol when reacts with HCN + $HCl/ZnCl_2$ followed by hydrolysis

(D) β-napthol when reacts with $H_3C-\overset{\overset{O}{\|}}{C}-Cl + AlCl_3$

AH0229

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EXERCISE # O-II

1. In which of the following reaction t-butylbenzene is formed:

(A) Benzene + iso-butyl chloride, $AlCl_3$

(B) Benzene + $(CH_3)_2C=CH_2 \xrightarrow{BF_3.HF}$

(C) Benzene + t-butyl alcohol $\xrightarrow{H_2SO_4}$

(D) Benzene + $(CH_3)_2C=CH_2 \xrightarrow{AlCl_3}$

AH0044

2. The replacement of a hydrogen atom in benzene by alkyl group can be brought about with the following reagents :

(A) Alkyl chloride and $AlCl_3$

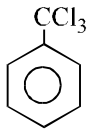
(B) Alkene and $AlCl_3$

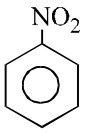
(C) Alkanol and alkali

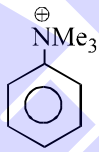
(D) Alkanol and acid

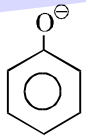
AH0045

3. Electrophilic attack of $NO_2^{\oplus}$ at meta position is observed in:

(A) 

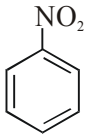
(B) 

(C) 

(D) 

AH0048

4. Which of the following can be used as reagent in Friedel Crafts reaction?

(A) 

(B) $CH_2=CH-Cl$

(C) CH_3CH_2Cl

(D) $CH_2=CH-CH_2-Cl$

AH0050

5. Of the species PhSR, $PhS(=O)R$, $PhS(=O)_2R$ and $Ph-S(=O)_2-OR$ the meta-substituted product is obtained from

(A) PhSR

(B) $PhS(=O)R$

(C) $PhS(=O)_2R$

(D) $Ph-S(=O)_2-OR$

AH0051

6. Amongst the following, the moderately activating group is

(A) —NHR

(B) $-NHCOCH_3$

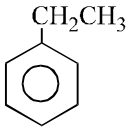
(C) $-O-C(=O)-R$

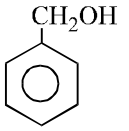
(D) $-CH_3$

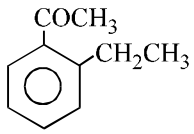
AH0052

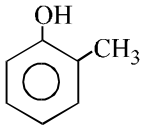
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\15-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

7. Benzoic acid may be prepared by the oxidation of:

(A) 

(B) 

(C) 

(D) 

AH0054

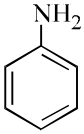
8. Which of the following is not an ortho-para directing group?

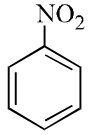
(A) CF_3 (B) CCl_3 (C) –CH=CH–COOH (D) $-N\equiv C$

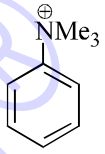
AH0058

9. Which of the following does not gives Fridel-Crafts reaction?

(A) 

(B) 

(C) 

(D) 

AH0059

10. Which of the following are classified as aromatic ?

(A) 1, 2, 3-Triphenylcyclopropenium cation (B) Cyclooctatetraenyl dianion
(C) Azulene (D) Annulene [10]

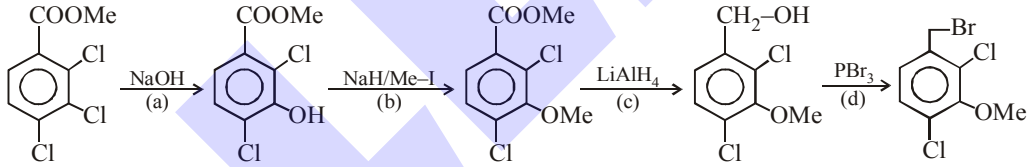
AH0061

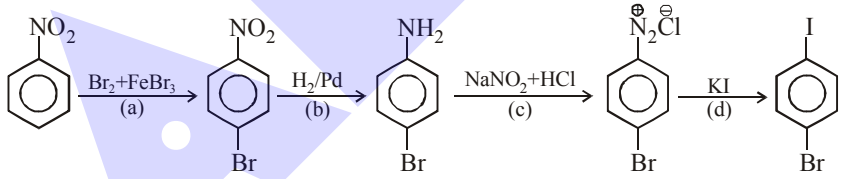
11. Which of the following is/are name of 1,2,3,4,5,6 - hexachloro cyclohexane :

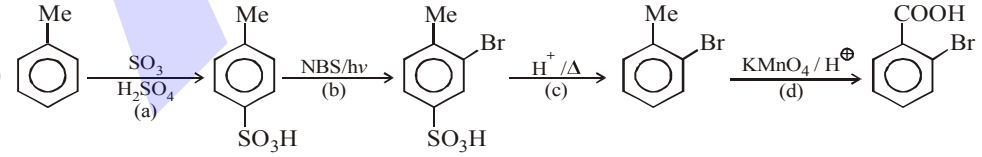
(A) Lindane (B) Gammexane (C) 666 (D) BHC

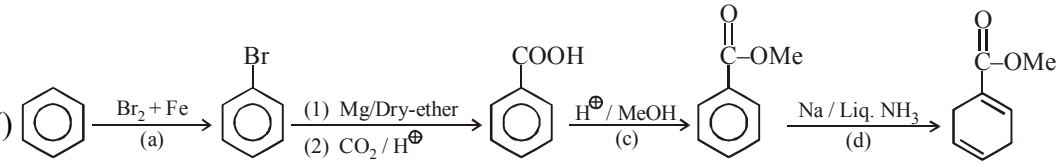
AH0062

12. Among the following reaction sequences identify incorrect step :

(I) 

(II) 

(III) 

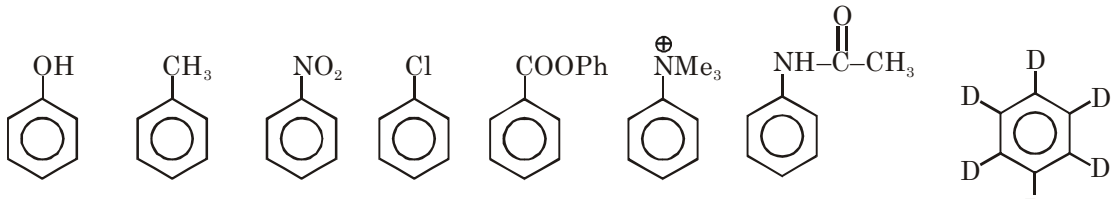
(IV) 

(A) I–a ; II – b ; III – c ; IV – d (B) I–b ; II – a ; III – d ; IV – d
(C) I–a ; II – a ; III – b ; IV – d (D) I–b ; II – a ; III – d ; IV – c

AH0063

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep, Tec-Che Every\Eng\2-Aromatic\01_Th & Ex.p65

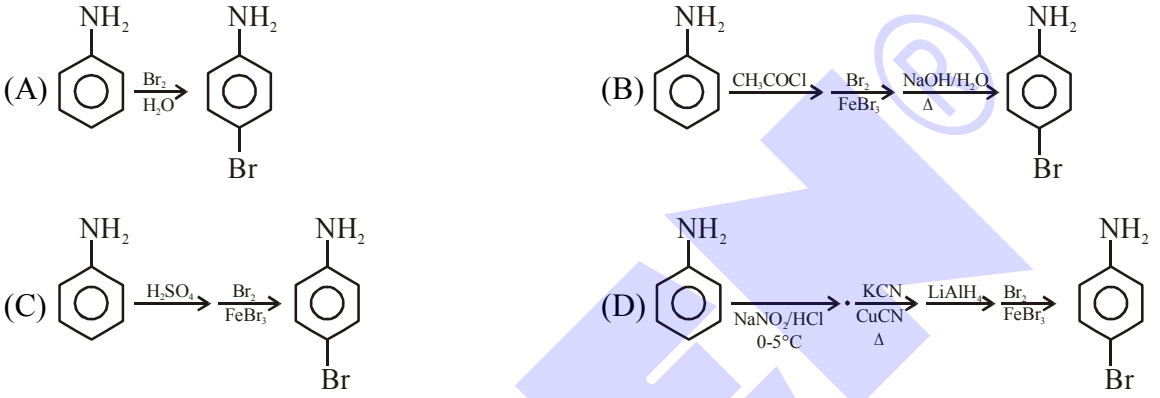
13. How many of following compounds are less reactive than benzene for sulphonation by conc. H_2SO_4 :



(A) 2 (B) 4 (C) 6 (D) 8

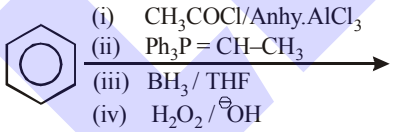
AH0064

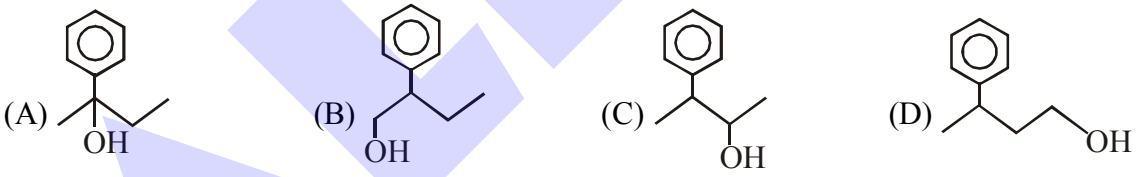
14. Which of the following method(s) is/are not used to prepare p-bromo aniline as major product :



AH0066

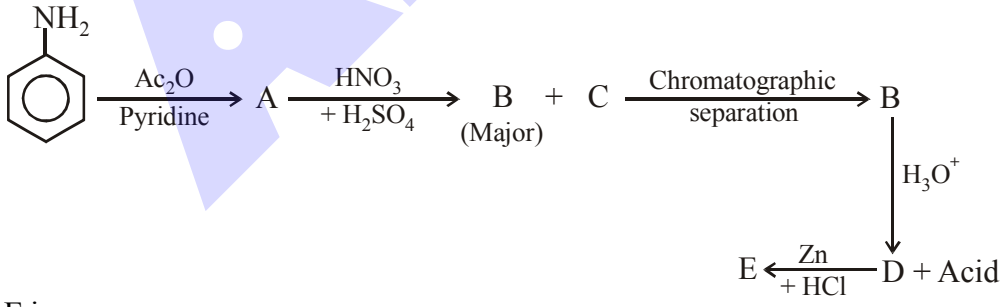
15. Which product is/are not obtained in following reaction.



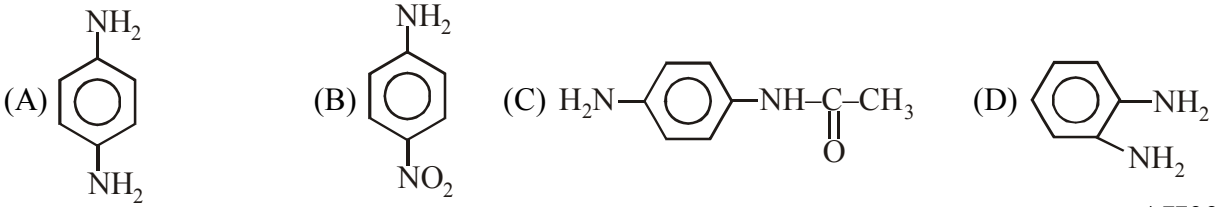


AH0068

16.



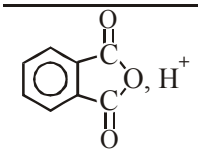
E is :



AH0069

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17. Cumene $\xrightarrow[h\nu]{O_2}$ A $\xrightarrow[H_2SO_4]{conc.}$ Acetone + B $\longrightarrow$ C.



Identify correct statement :

(A) Product C is phenophthalein indicater

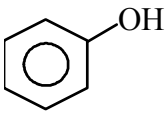
(B) Product B gives CO_2 effervescence with $NaHCO_3$

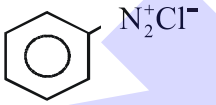
(C) Product A formation involves carbocation intermediate

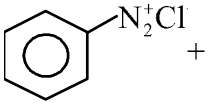
(D) Product B gives no colour with neutral $FeCl_3$

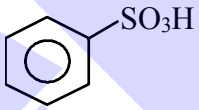
AH0070

18. Which of the following reactions gives mixture of benzene & dibasic acid as product ?

(A)  $+ Zn \rightarrow$

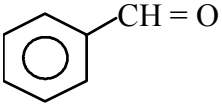
(B)  $+ H_3PO_2 + H_2O \rightarrow$

(C)  $+ CH_3CH_2OH \rightarrow$

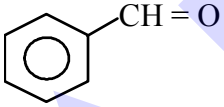
(D)  + dilute acid $\rightarrow$

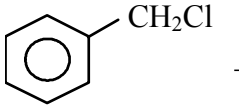
AH0230

19. Which of the following reactions will give toluene by reduction of given compounds:

(A)  $\xrightarrow{H_2(Ni)}$

(B) n-heptane $\xrightarrow{Cr_2O_3/Al_2O_3/MoO_2/\Delta}$

(C)  $\xrightarrow{N_2H_4+OH^-/\Delta}$

(D)  $\xrightarrow{LiAlH_4}$

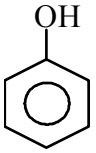
AH0231

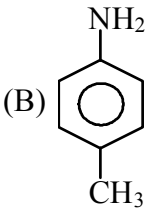
20. Which reagent reacts with aniline to give water insoluble product ?

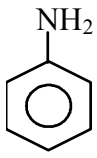
(A) $NaNO_2 + HBF_4$ (B) $NaNO_2 + HCl$ (C) $K_2Cr_2O_7$ (D) CF_3CO_3H

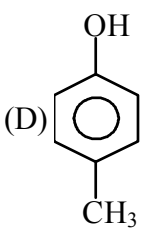
AH0232

21. p-toluidine when reacts with benzene diazonium chloride to form a compound which on boiling with H_2SO_4 give

(A) 

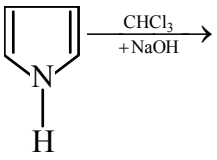
(B) 

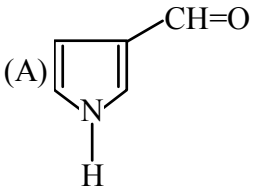
(C) 

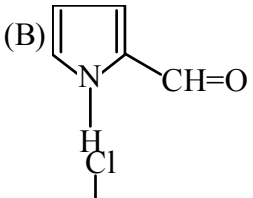
(D) 

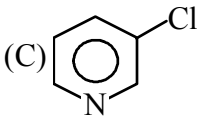
AH0233

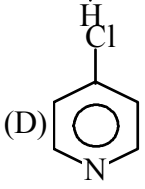
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep, Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex..p65

22. The possible products in following reaction 

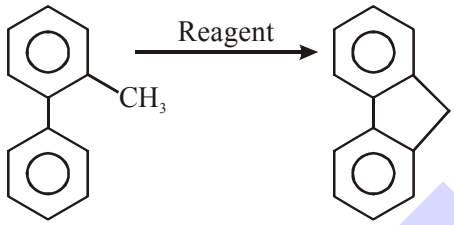
(A) 

(B) 

(C) 

(D) 

AH0234

23. 

The sequence of reagents can be

(A) SO_2Cl_2, $AlCl_3$

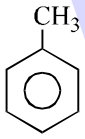
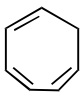
(B) $Cl_2/AlCl_3$; Na(D.E.)

(C) $SOCl_2$, $AlCl_3$

(D) NBS, $FeBr_3$

AH0235

24. Match the column :

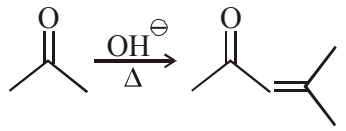
Column I (Reaction)		**Column II (Product & its property)**	
(A)	 + CrO_3 in $CH_3COOH \longrightarrow$?	(P)	Tropylium ion
(B)	 + $Ph_3C^{\oplus} \longrightarrow$?	(Q)	Benzaldehyde
(C)	 + n-Pr Br + $ZnCl_2 \longrightarrow$?	(R)	Product can oxidise by $KMnO_4/H^+$
(D)	C_6H_6 + CO + HCl $\xrightarrow{AlCl_3}$?	(S)	Aromatic product obtained

AH0075

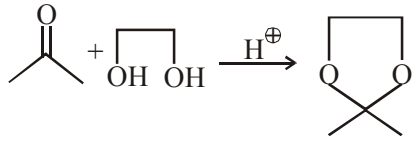
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biamole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

25. **Column - I** **Column - II**

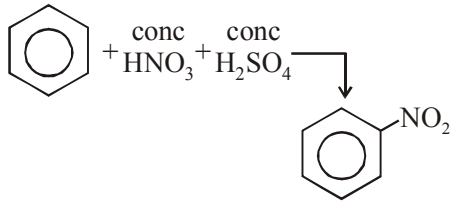
(Reactions **(Intermediate formed or type of reaction)**

(A) 

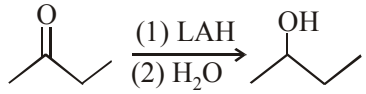
(P) Product obtained as racemic mixture

(B) 

(Q) Substitution reaction

(C) 

(R) $Nu^{\ominus}$ Addition takes place during reaction

(D) 

(S) Carbocation intermediate

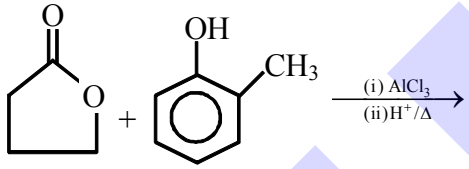
(T) Carbanion intermediate

AH0076

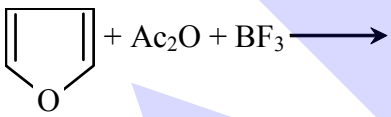
26. **Match the column :**

Column I **Column II**

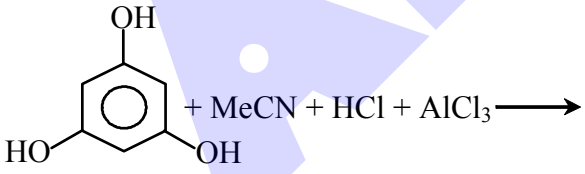
(Reaction) **(Product & its property)**

(A) 

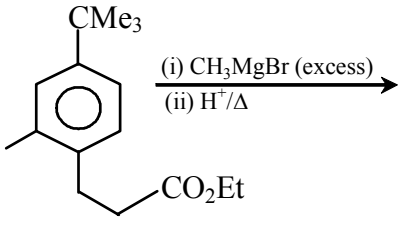
(P) Acyl group in introduced in ring.

(B) 

(Q) Intramolecular electrophilic aromatic substitution

(C) 

(R) Alkylation takes place

(D) 

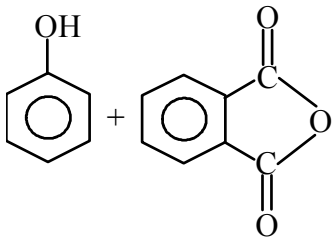
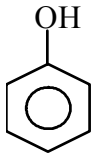
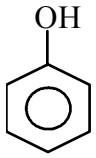
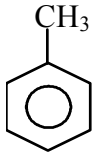
(S) Product gives +ve test with 2, 4, D.N.P. solution

(T) Aromatic hydrocarbon is formed as product

AH0236

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27. Match the column :

Column I (Reaction)		**Column II (Product & its property)**	
(A)	 + conc. $H_2SO_4 \rightarrow$	(P)	Electrophilic aromatic substitution
(B)	 + HCHO (Basic solution) $\rightarrow$	(Q)	Liber mann-nitroso test
(C)	 + $NaNO_2$ + dil. $H_2SO_4 \rightarrow$	(R)	Explosive
(D)	 + fuming HNO_3(Xs) $\rightarrow$	(S)	Phthalein test
		(T)	Synthetic polymer

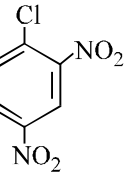
AH0237

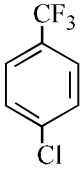
ALLEN®

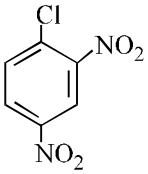
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

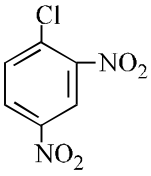
EXERCISE # S-I

1. Write the principal organic product in each of the following reactions:

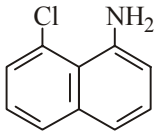
(i)  $\xrightarrow[\text{triethylene glycol}]{H_2NNH_2}$ B

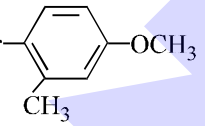
(ii)  $\xrightarrow[(ii)\,NaOCH_3,CH_3OH]{(i)\,HNO_3,H_2SO_4}$ C

(iii)  $+ CH_3CH_2ONa \longrightarrow$ D

(iv)  $+ C_6H_5CH_2SNa \longrightarrow$ E

AH0078

2. (i)  $\xrightarrow[(ii)\,CuBr]{(i)\,NaNO_2,HBr}$ E

(ii)  $\xrightarrow[(ii)\,NaSCH_3]{(i)\,NBS,\text{Benzoyl peroxide},CCl_4,\text{heat}}$ F

AH0079

3. Compare the given characteristics of aniline and cyclohexanamine :

(i) Both are primary amine

(ii) Both can be acylated by RCOCl

(iii) Both reacts with $CHCl_3$/KOH

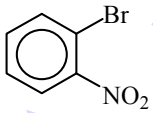
(iv) Both reacts with $NaNO_2$ + HCl at 0-5°C

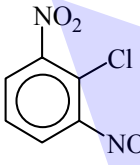
(v) Both reacts with $PhSO_2Cl$ to give a compound which is soluble in KOH

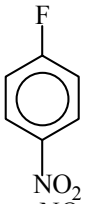
(vi) Both gives coupling reaction with phenol

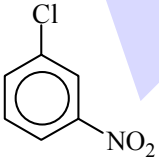
(vii) Both gives electrophilic substitution reaction

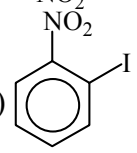
How many of the given characteristics are correct? **AH0081**

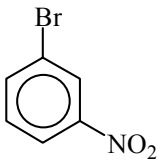
4. Number of compounds which can show faster rate of nucleophilic substitution of halogen than  :

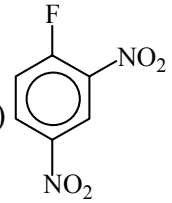
(i) 

(ii) 

(iii) 

(iv) 

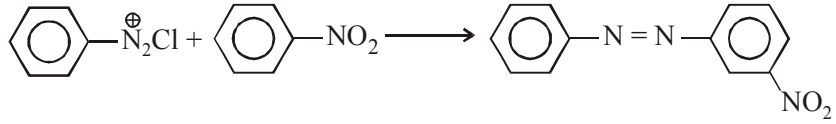
(v) 

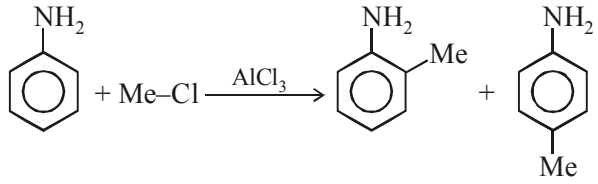
(vi) 

AH0082

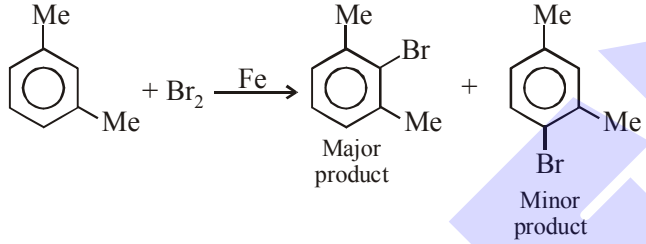
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

5. Identify total number of reactions incorrectly match with its product ?

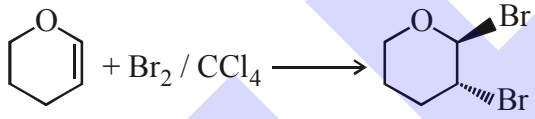
(a) 

(b) 

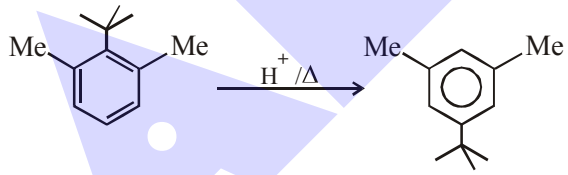
(c) $Et–OH + NaI \xrightarrow[H_2SO_4]{conc.} Et - I$

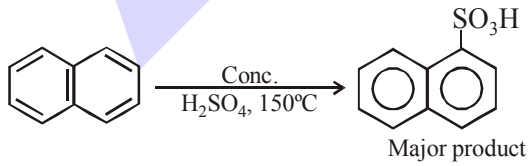
(d) 

(e) $Me–C\equiv CH + Na/Liq\ NH_3 \longrightarrow Me–CH = CH_2$

(f) 

100 % product [Stereo specific reactions]

(g) 

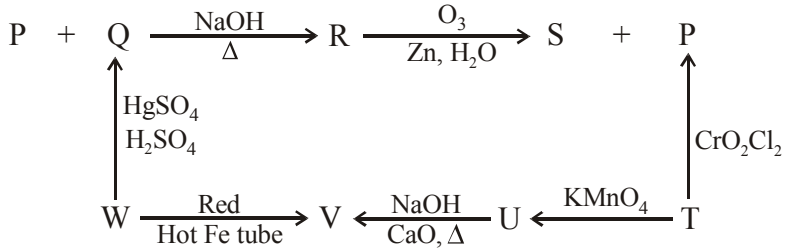
(h) 

AH0083

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\15-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

Paragraph for No. 6 to 7

For given reaction sequence molecular formula for compound 'U' is $C_7H_6O_2$ & P gives negative Fehling test.

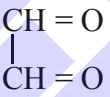
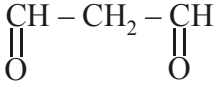


6. Compound which is not a hydrocarbon

(A) W (B) R (C) T (D) V

AH0084

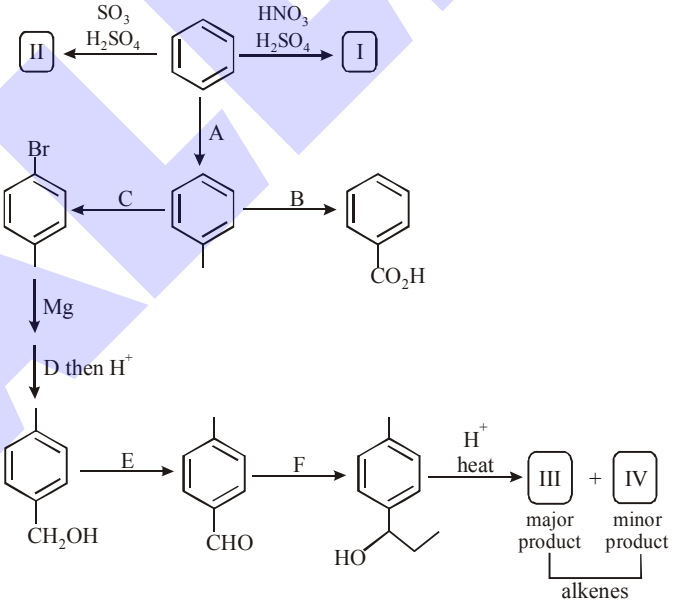
7. Compound S is :

(A) $CH_3 - CH = O$ (B) Ph – CH = O (C)  (D) 

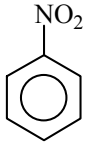
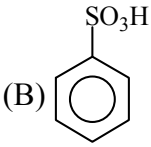
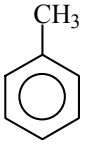
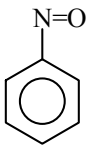
AH0084

Paragraph for 8 to 9

Identify reagent used and intermediate products in following conversion.



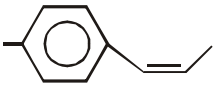
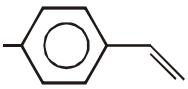
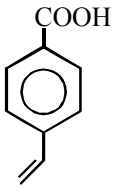
8. Identify II product -

(A)  (B)  (C)  (D) 

AH0072

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic Aromatic Biomole-Poly, POC & Sep, Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex..p65

9. (III) major product is ?

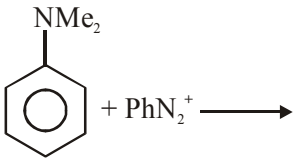
(A)  (B)  (C)  (D) 

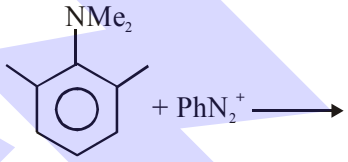
AH0072

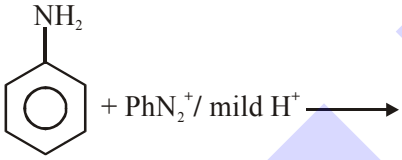
10. An oily liquid (A) is insoluble in H_2O but on heating with aqueous NaOH for one hour, it get dissolved. From the reaction mixture, a liquid (B) is distilled which gives a yellow ppt with NaOI. On careful oxidation of (B) it gives an aldehyde (C) which also gives positive iodoform test. If H_2SO_4 is added to the remaining solution then a white crystal (D) is formed which gives +ve test with $NaHCO_3$. Heating (D) with NaOH + CaO gives benzene. Find the minimum molecular weight of (A).

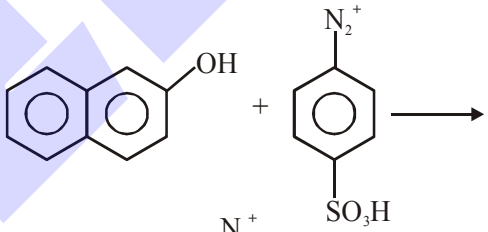
AH0238

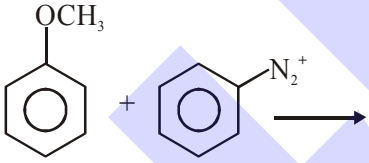
11. How many reactions do not give azo dye when react with benzene diazonium ion?

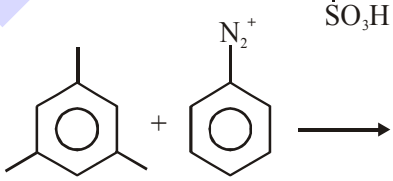
(a) 

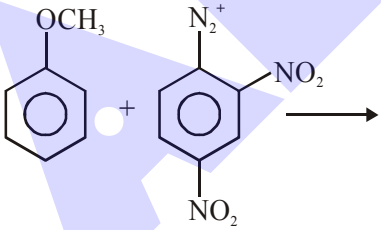
(b) 

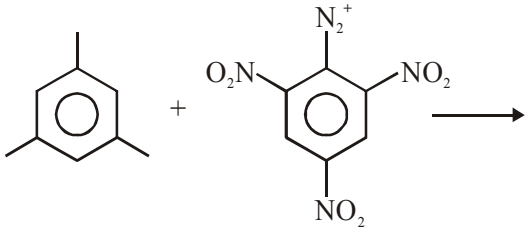
(c) 

(d) 

(e) 

(f) 

(g) 

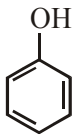
(h) 

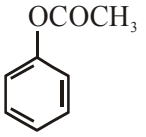
AH0239

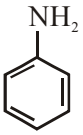
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

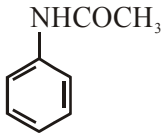
EXERCISE # (MAINS)

1. Which of the following compounds will significant amont of meta product during mono-nitration reaction ?
[JEE(Main)-2017]

(1) 

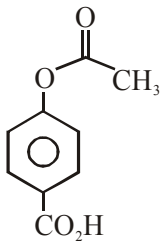
(2) 

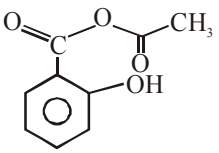
(3) 

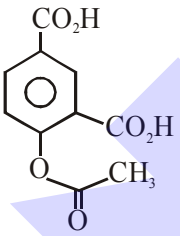
(4) 

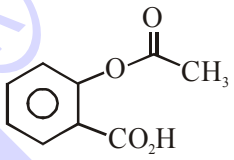
AH0092

2. Phenol on treatment with CO_2 in the presence of NaOH followed by acidification produces compound X as the major product. X on treatment with $(CH_3CO)_2O$ in the presence of catalytic amount of H_2SO_4 produces :
[JEE(Main)-2018]

(1) 

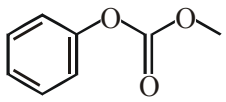
(2) 

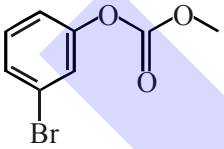
(3) 

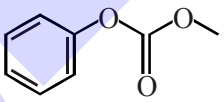
(4) 

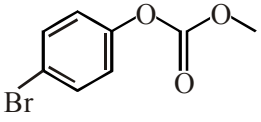
AH0093

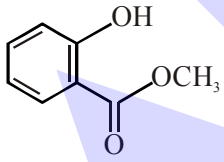
3. Phenol reacts with methyl chloroformate in the presence of NaOH to form product A. A reacts with Br_2 to form product B. A and B are respectively :
[JEE(Main)-2018]

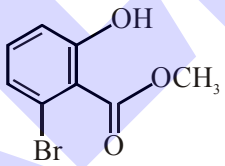
(1) 

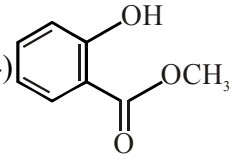
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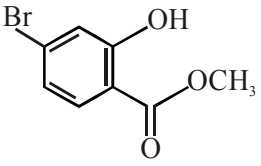
(2) 

and 

(3) 

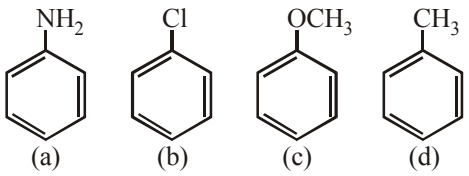
and 

(4) 

and 

AH0094

4. The increasing order of nitration of the following compound is :- **[JEE(Main)-2018(ONLINE)]**

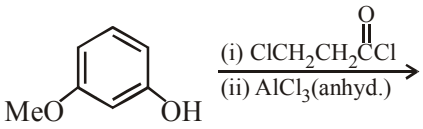


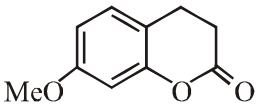
(1) (b) < (a) < (c) < (d)
(2) (b) < (a) < (d) < (c)
(3) (a) < (b) < (c) < (d)
(4) (a) < (b) < (d) < (c)

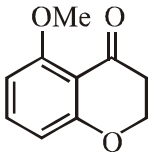
AH0095

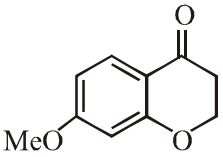
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

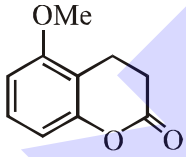
5. The major product of the following reaction is : **[JEE(Main)-2018(ONLINE)]**



(1) 

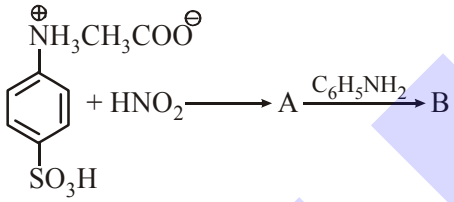
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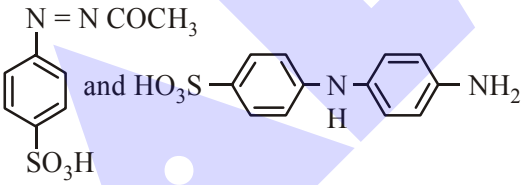
(3) 

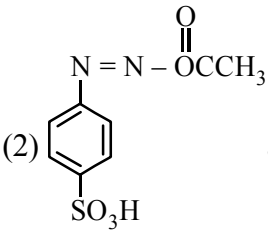
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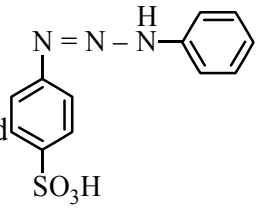
AH0096

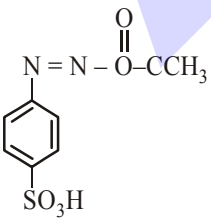
6. Products A and B formed in the following reactions are respectively : **[JEE(Main)-2018(ONLINE)]**

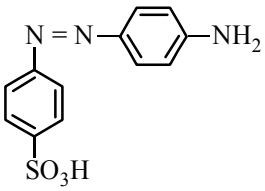


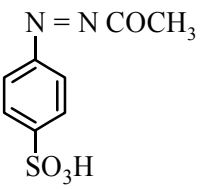
(1) 

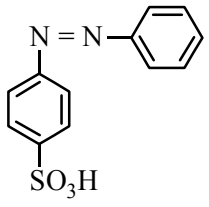
(2) 

and 

(3) 

and 

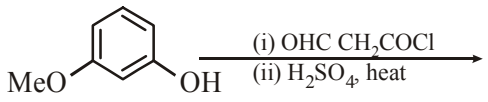
(4) 

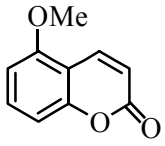
and 

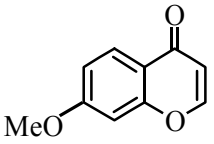
AH0097

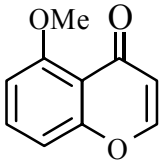
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

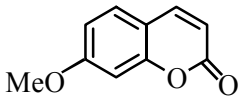
7. The major product of the following reaction is : **[JEE(Main)-2018(ONLINE)]**



(1) 

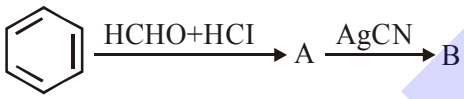
(2) 

(3) 

(4) 

AH0098

8. The compounds A and B in the following reaction are, respectively: **[JEE-Mains (JAN)-2019]**



(1) A = Benzyl alcohol, B = Benzyl isocyanide

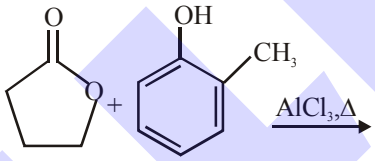
(2) A = Benzyl alcohol, B = Benzyl cyanide

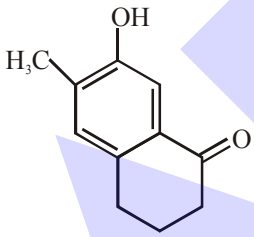
(3) A = Benzyl chloride, B = Benzyl cyanide

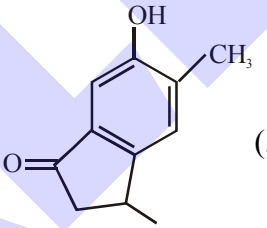
(4) A = Benzyl chloride, B = Benzyl isocyanide

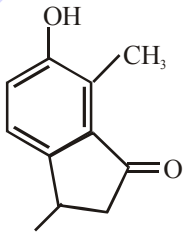
AH0099

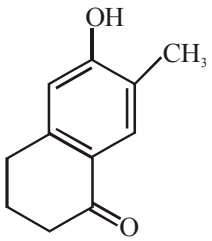
9. The major product of the following reaction is: **[JEE-Mains (JAN)-2019]**



(1) 

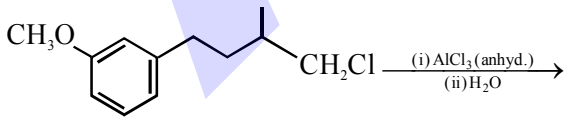
(2) 

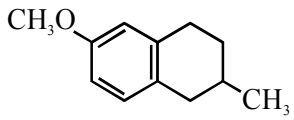
(3) 

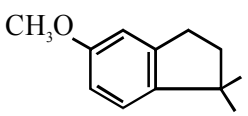
(4) 

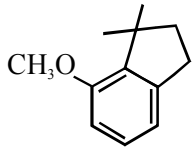
AH0100

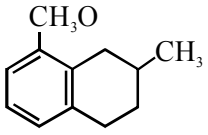
10. The major product of the following reaction is: **[JEE-Mains (JAN)-2019]**



(1) 

(2) 

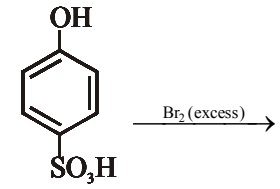
(3) 

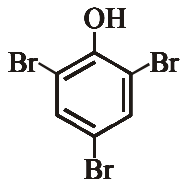
(4) 

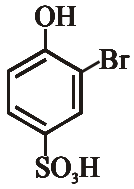
AH0102

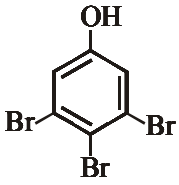
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep, Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

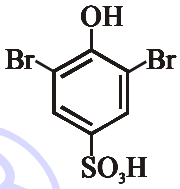
11. The major product of the following reaction is : **[JEE-Mains (JAN)-2019]**



(1) 

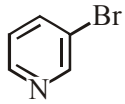
(2) 

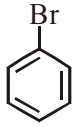
(3) 

(4) 

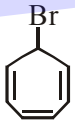
AH0105

12. Which of the following compounds will produce a precipitate with $AgNO_3$? **[JEE-Mains (JAN)-2019]**

(1) 

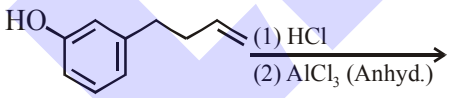
(2) 

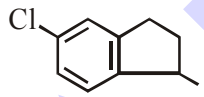
(3) 

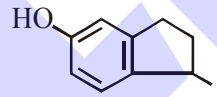
(4) 

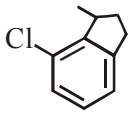
AH0106

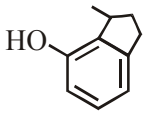
13. The major product of the following reactionis :- **[JEE-Mains (JAN)-2019]**



(1) 

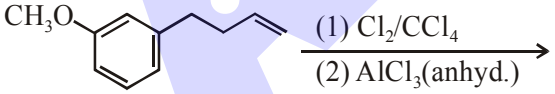
(2) 

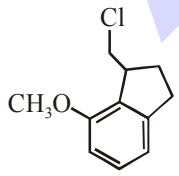
(3) 

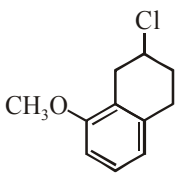
(4) 

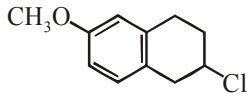
AH0107

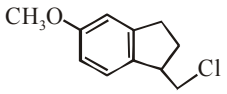
14. The major product of the following reaction is : **[JEE-Mains (JAN)-2019]**



(1) 

(2) 

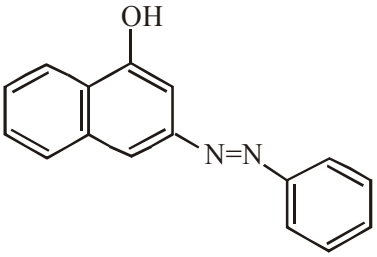
(3) 

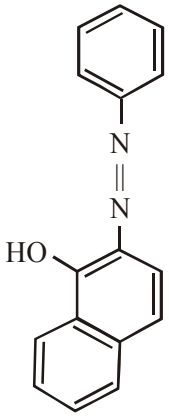
(4) 

AH0109

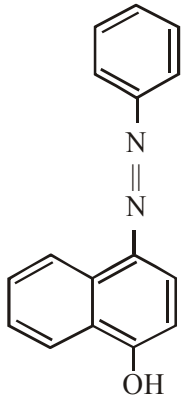
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

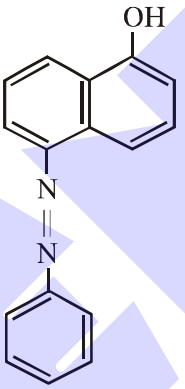
15. Coupling of benzene diazonium chloride with 1-napthol in alkaline medium will give

(1) 

(2) 

[JEE-Mains (April)-2019]

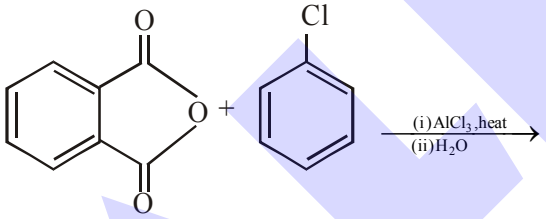
(3) 

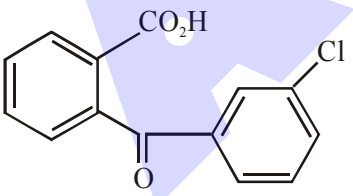
(4) 

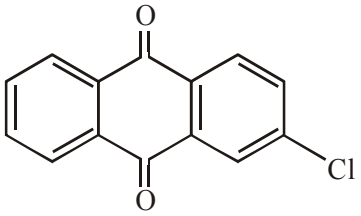
AH0111

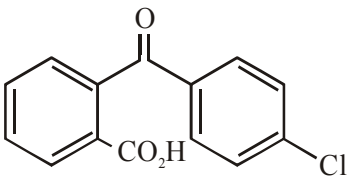
16. The major product of the following reaction is:

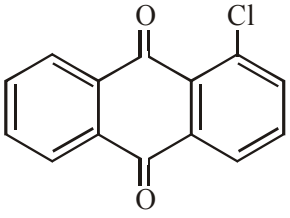
[JEE-Mains (April)-2019]



(1) 

(2) 

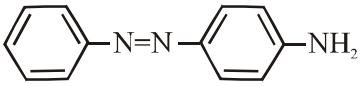
(3) 

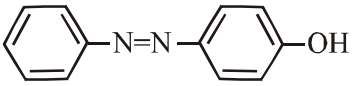
(4) 

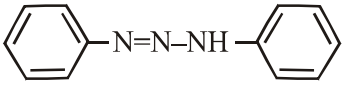
AH0112

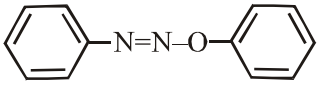
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\5-Carboxylic Aromatic Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

17. Aniline dissolved in dilute HCl is reacted with sodium nitrite at 0ºC. This solution was added dropwise to a solution containing equimolar mixture of aniline and phenol in dil. HCl. The structure of the major product is : **[JEE-Mains (April)-2019]**

(1) 

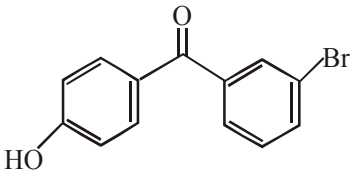
(2) 

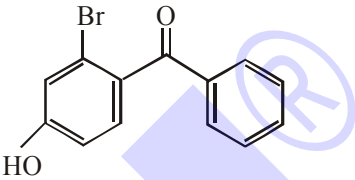
(3) 

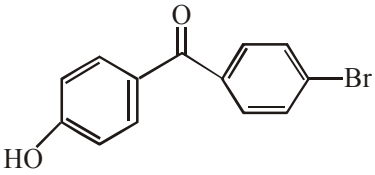
(4) 

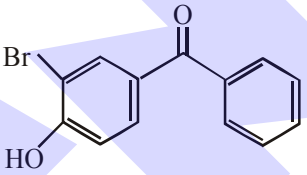
AH0115

18. p-Hydroxybenzophenone upon reaction with bromine in carbon tetrachloride gives: **[JEE-Mains (April)-2019]**

(1) 

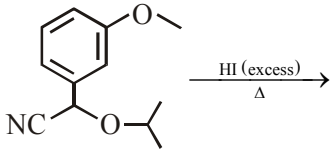
(2) 

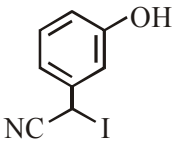
(3) 

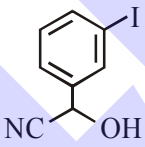
(4) 

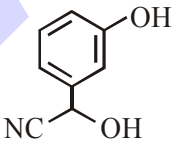
AH0116

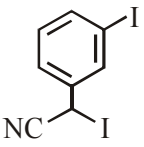
19. The major product of the following reaction is : **[JEE-Mains (April)-2019]**



(1) 

(2) 

(3) 

(4) 

AH0117

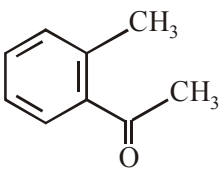
20. Which of the following is NOT a correct method of the preparation of benzylamine from cyanobenzene ? **[JEE-Mains (April)-2019]**

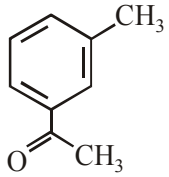
(1) (i) HCl/H_2O (ii) $NaBH_4$
(2) (i) $LiAlH_4$ (ii) H_3O^+
(3) (i) $SnCl_2$+HCl(gas) (ii) $NaBH_4$
(4) H_2/Ni

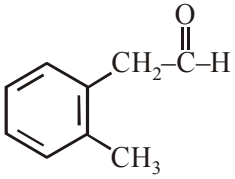
AH0119

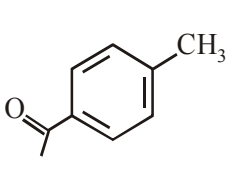
21. Compound A ($C_9H_{10}O$) shows positive iodoform test. Oxidation of A with $KMnO_4$/KOH gives acid B($C_8H_6O_4$). Anhydride of B is used for the preparation of phenolphthalein. Compound A is :-

[JEE-Mains (April)-2019]

(1) 

(2) 

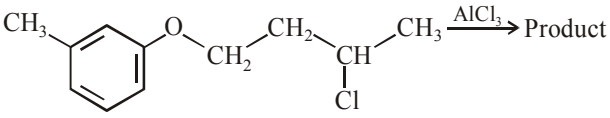
(3) 

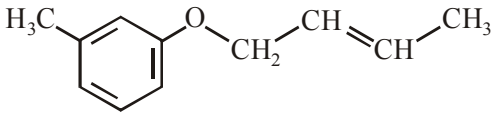
(4) 

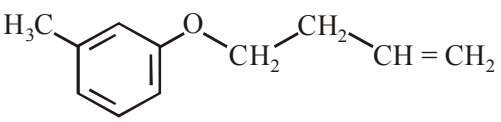
AH0120

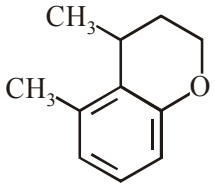
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

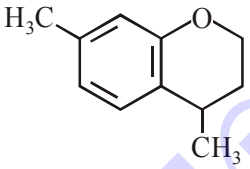
22. The major product obtained in the given reaction is :- **[JEE-Mains (April)-2019]**



(1) 

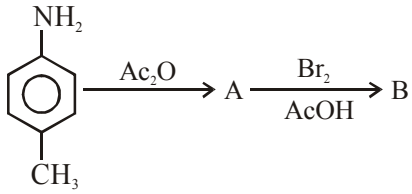
(2) 

(3) 

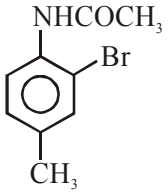
(4) 

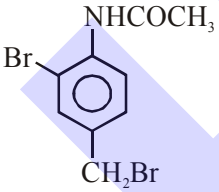
AH0121

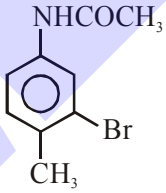
23. In the following reaction sequence **[JEE-Mains (Jan)-2020]**

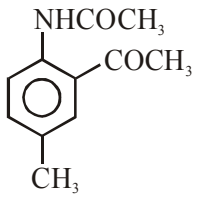


the major products B is -

(1) 

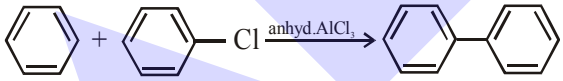
(2) 

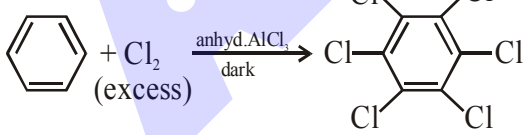
(3) 

(4) 

AH0122

24. Consider the following reactions : **[JEE-Mains (Jan)-2020]**

(a) 

(b) 

(c) 

(d) 

Which of these reactions are possible ?

(1) (a) and (d) (2) (b) and (d) (3) (a) and (b) (4) (b) , (c) and (d)

AH0123

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25. A solution of m-chloroaniline, m-chlorophenol and m-chlorobenzoic acid in ethyl acetate was extracted initially with a saturated solution of $NaHCO_3$ to give fraction A. The left over organic phase was extracted with dilute NaOH solution to give fraction B. The final organic layer was labelled as fraction C. Fractions A, B and C, contain respectively : **[JEE-Mains (Jan)-2020]**

(1) m-chlorobenzoic acid, m-chloroaniline and m-chlorophenol

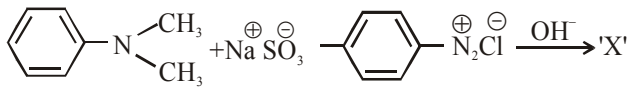
(2) m-chloroaniline, m-chlorobenzoic acid and m-chlorophenol

(3) m-chlorobenzoic acid, m-chlorophenol and m-chloroaniline

(4) m-chlorophenol, m-chlorobenzoic acid and m-chloroaniline

AH0124

26. Consider the following reaction : **[JEE-Mains (Jan)-2020]**



The product 'X' is used :

(1) in acid base titration as an indicator

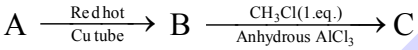
(2) in protein estimation as an alternative to ninhydrin

(3) in laboratory test for phenols

(4) as food grade colourant

AH0125

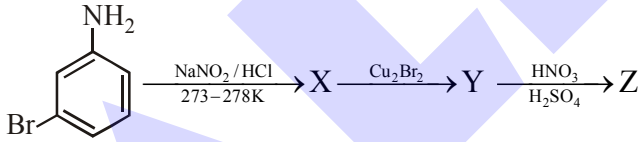
27. In the following sequence of reactions the maximum number of atoms present in molecule 'C' in one plane is _____. **[JEE-Mains (Jan)-2020]**

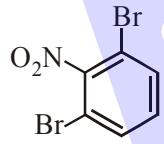


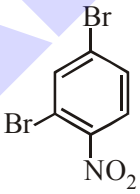
(A is a lowest molecular weight alkyne)

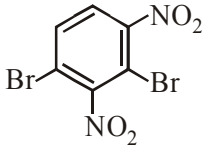
AH0126

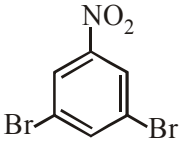
28. The major product Z obtained in the following reaction scheme is : **[JEE-Mains (sep)-2020]**



(1) 

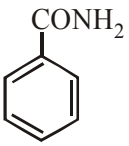
(2) 

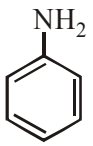
(3) 

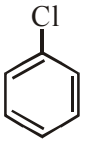
(4) 

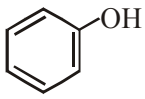
AH0183

29. Which of these will produce the highest yield in Friedel Crafts reaction? **[JEE-Mains (sep)-2020]**

(1) 

(2) 

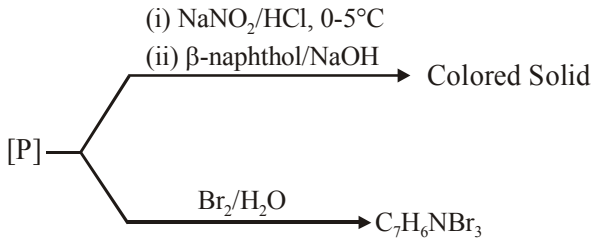
(3) 

(4) 

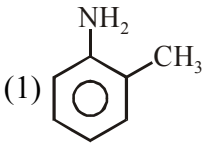
AH0184

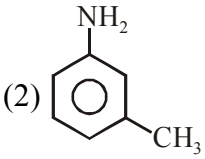
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

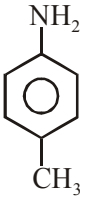
30. Consider the following reactions, **[JEE-Mains (sep)-2020]**

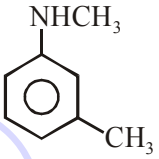


The compound [P] is :

(1) 

(2) 

(3) 

(4) 

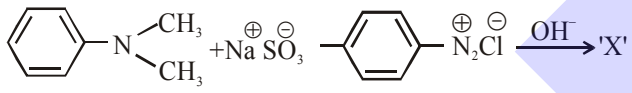
AH0185

31. The number of sp^2 hybrid orbitals in a molecule of benzene is : **[JEE-Mains (sep)-2020]**

(1) 24 (2) 6 (3) 12 (4) 18

AH0186

32. Consider the following reaction : **[JEE-Mains (sep)-2020]**



The product 'X' is used :

(1) in acid base titration as an indicator

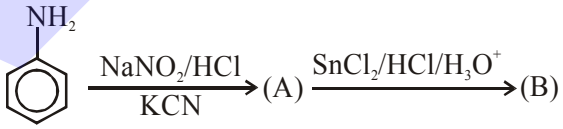
(2) in protein estimation as an alternative to ninhydrin

(3) in laboratory test for phenols

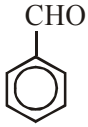
(4) as food grade colourant

AH0187

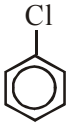
33. 'A' and 'B' in the following reactions are : **[JEE-Mains-2021]**

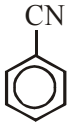


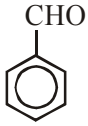
(1) (A) : 

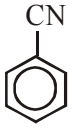
(B) : 

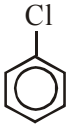
(2) (A) : 

(B) : 

(3) (A) : 

(B) : 

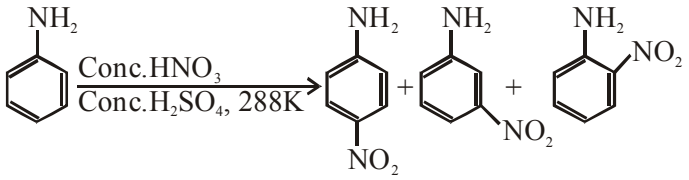
(4) (A) : 

(B) : 

AH0188

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

34. In the following reaction the reason why meta-nitro product also formed is : **[JEE-Mains-2021]**



(1) low temperature

(2) $-NH_2$ group is highly meta-directive

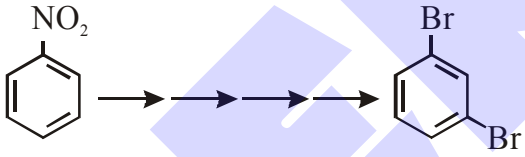
(3) Formation of anilinium ion

(4) $-NO_2$ substitution always takes place at meta-position

AH0189

35. What is the correct sequence of reagents used for converting nitrobenzene into *m*-dibromobenzene ?

[JEE-Mains-2021]



(1) $\xrightarrow{NaNO_2}/\xrightarrow{HCl}/\xrightarrow{KBr}/\xrightarrow{H^+}$

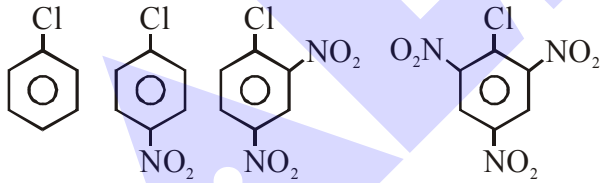
(2) $\xrightarrow{Br_2/Fe}/\xrightarrow{Sn/HCl}/\xrightarrow{NaNO_2/HCl}/\xrightarrow{CuBr/HBr}$

(3) $\xrightarrow{Sn/HCl}/\xrightarrow{KBr}/\xrightarrow{Br_2}/\xrightarrow{H^+}$

(4) $\xrightarrow{Sn/HCl}/\xrightarrow{Br_2}/\xrightarrow{NaNO_2}/\xrightarrow{NaBr}$

AH0190

36. The correct order of the following compounds showing increasing tendency towards nucleophilic substitution reaction is :-



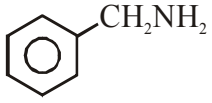
[JEE-Mains-2021]

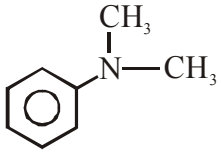
(1) (iv) < (iii) < (ii) < (i) (2) (iv) < (i) < (ii) < (iii)

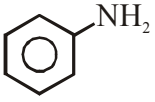
(3) (iv) < (i) < (iii) < (ii) (4) (i) < (ii) < (iii) < (iv)

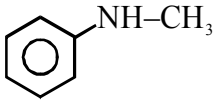
AH0191

37. The diazonium salt of which of the following compounds will form a coloured dye on reaction with β-Naphthol in NaOH ? **[JEE-Mains-2021]**

(1) 

(2) 

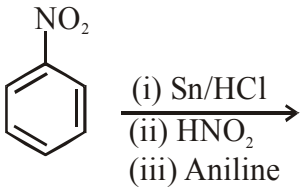
(3) 

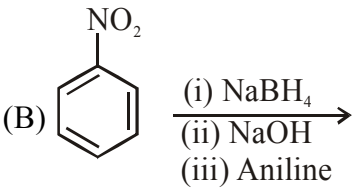
(4) 

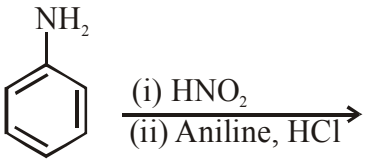
AH0192

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38. Which of the following reaction/s will not give p-aminoazobenzene? **[JEE-Mains-2021]**

(A) 

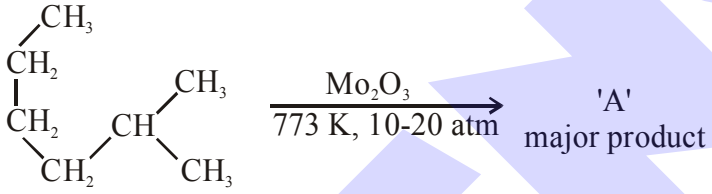
(B) 

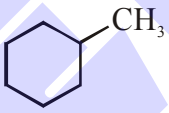
(C) 

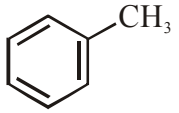
(1) A only (2) B only (3) C only (4) A and B

AH0193

39. Identify A in the given chemical reaction. **[JEE-Mains-2021]**

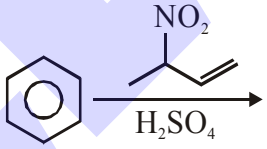


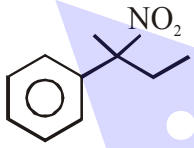
(1)  (2)  (3) 

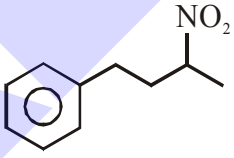
(4) 

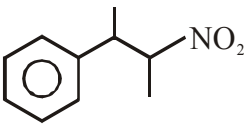
AH0194

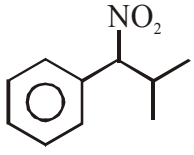
40. The major product of the following reaction is: **[JEE-Mains-2021]**



(1) 

(2) 

(3) 

(4) 

AH0195

41. The correct sequence of reagents used in the preparation of 4-bromo-2-nitroethyl benzene from benzene is : **[JEE-Mains-2021]**

(1) HNO_3/H_2SO_4, $Br_2/AlCl_3$, $CH_3COCl/AlCl_3$, Zn-Hg/HCl

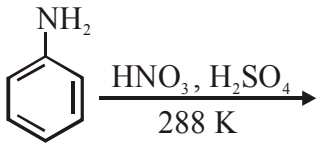
(2) $Br_2/AlBr_3$, $CH_3COCl/AlCl_3$, HNO_3/H_2SO_4, Zn/HCl

(3) $CH_3COCl/AlCl_3$, $Br_2/AlBr_3$, HNO_3/H_2SO_4, Zn/HCl

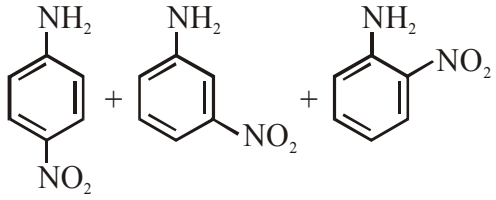
(4) $CH_3COCl/AlCl_3$, Zn-Hg/HCl, $Br_2/AlBr_3$, HNO_3/H_2SO_4

AH0196

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42. 

[JEE-Mains-2021]



(A) (B) (C)

Correct statement about the given chemical reaction is :

(1) $-\ddot{N}H_2$ group is *ortho* and *para* directive, so product (B) is not possible.

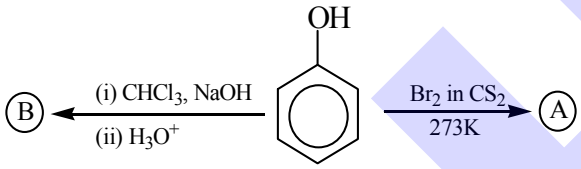
(2) Reaction is possible and compound (B) will be the major product.

(3) The reaction will form sulphonated product instead of nitration.

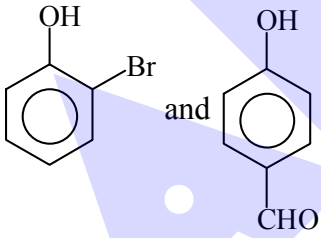
(4) Reaction is possible and compound (A) will be major product.

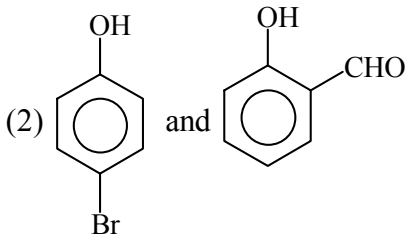
AH0197

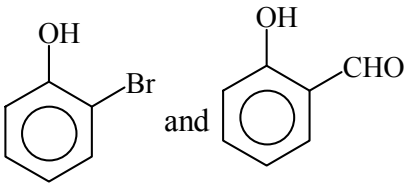
43. Identify the major products A and B respectively in the following reactions of phenol.

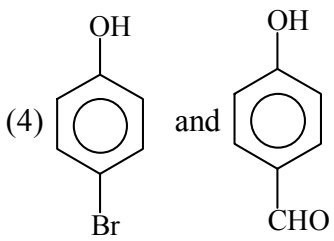


[JEE-Mains-2021]

(1) 

(2) 

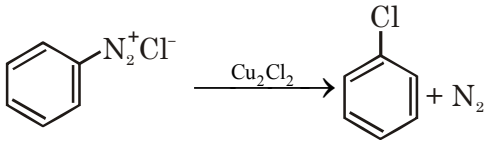
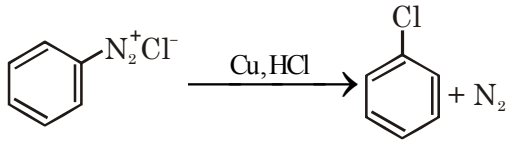
(3) 

(4) 

AH0198

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44. Match List-I with List-II **[JEE-Mains-2021]**

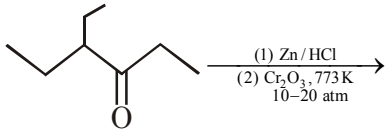
List-I	**List-II**
(a) 	(i) Wurtz reaction
(b) 	(ii) Sandmeyer reaction
(c) $2CH_3CH_2Cl + 2Na \xrightarrow{Ether} C_2H_5–C_2H_5+2NaCl$	(iii) Fittig reaction
(d) $2C_6H_5Cl + 2Na \xrightarrow{Ether} C_6H_5–C_6H_5 + 2NaCl$	(iv) Gatterman reaction

Choose the correct answer from the options given below :

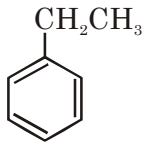
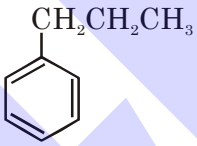
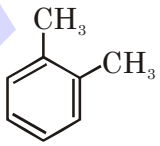
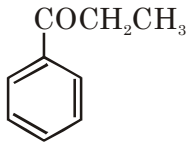
(1) (a) → (iii), (b) → (i), (c) → (iv), (d) → (ii) (2) (a) → (ii), (b) → (i), (c) → (iv), (d) → (iii)

(3) (a) → (ii), (b) → (iv), (c) → (i), (d) → (iii) (4) (a) → (iii), (b) → (iv), (c) → (i), (d) → (ii)

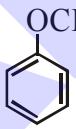
AH0199

45.  **[JEE-Mains-2021]**

considering the above reaction, the major product among the following is :

(1)  (2)  (3)  (4) 

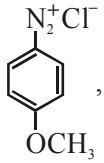
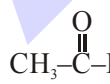
AH0200

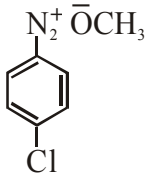
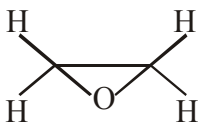
46. $C_7H_7N_2OCl + C_2H_5OH \rightarrow$  $+ N_2 +$ "X" + "Y"

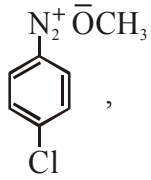
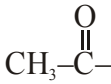
(A)

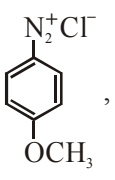
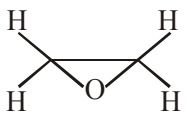
[JEE-Mains-2021]

In the above reaction, the structural formula of (A), "X" and "Y" respectively are :

(1)  ,  , HCl

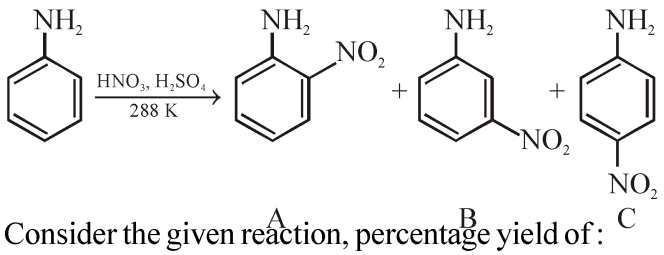
(2)  ,  , HCl

(3)  ,  , H_2O

(4)  ,  , H_2O

AH0202

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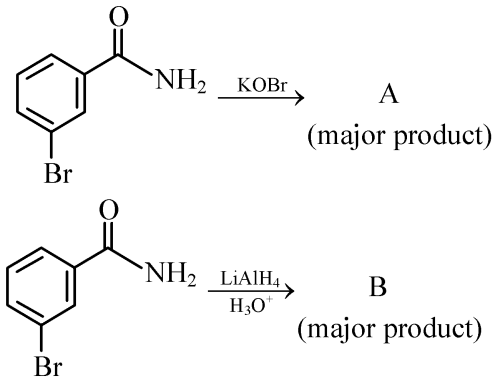
47. 

[JEE-Mains-2021]

Consider the given reaction, percentage yield of :

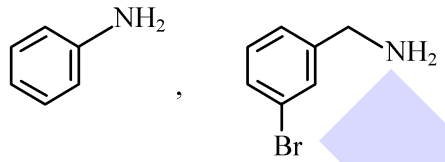
(1) C > A > B (2) B > C > A (3) A > C > B (4) C > B > A

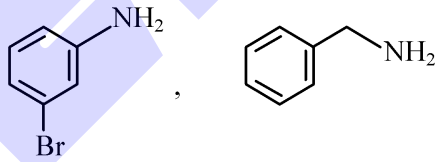
AH0204

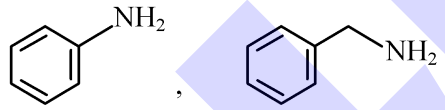
48. 

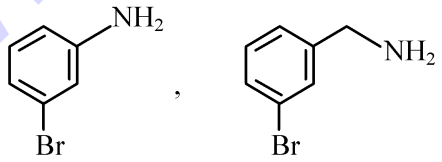
[JEE-Mains-2021]

In the above reactions, product A and product B respectively are :

(1) 

(2) 

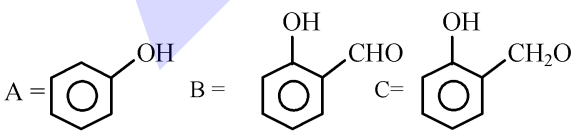
(3) 

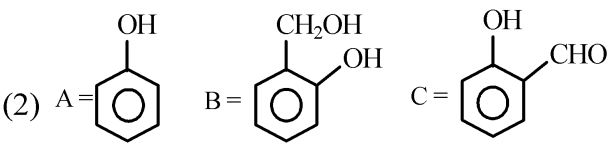
(4) 

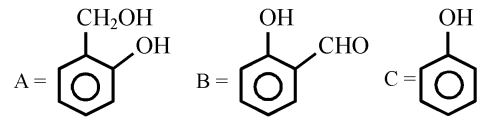
AH0205

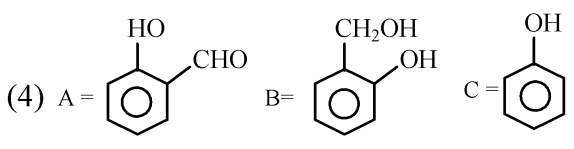
49. An organic compound A (C_6H_6O) gives dark green colouration with ferric chloride. On treatment with $CHCl_3$ and KOH, followed by acidification gives compound B. Compound B can also be obtained from compound C on reaction with pyridinium chlorochromate (PCC). Identify A, B and C .

[JEE-Mains-2021]

(1) 

(2) 

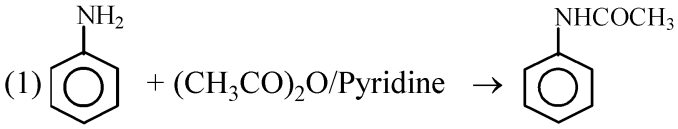
(3) 

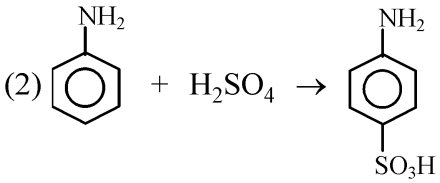
(4) 

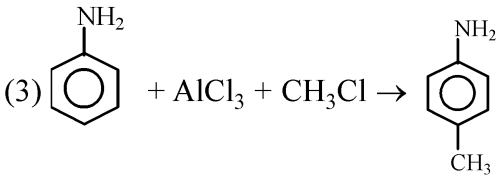
AH0208

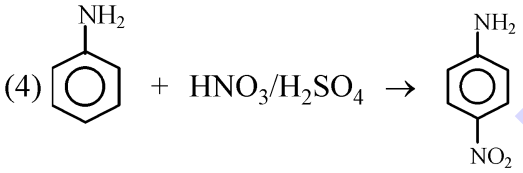
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep, Tec-Che Every day\Eng\2-Aromatic\01_TH & Ex.p65

50. Which one of the following reactions does not occur ? **[JEE-Mains-2021]**

(1) 

(2) 

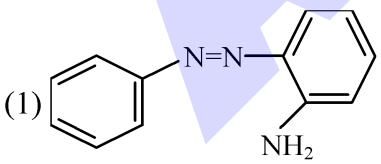
(3) 

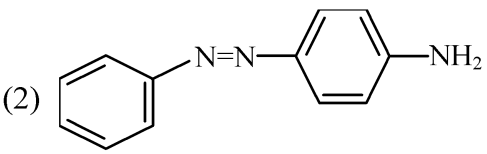
(4) 

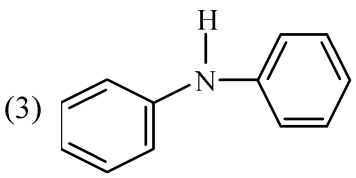
AH0209

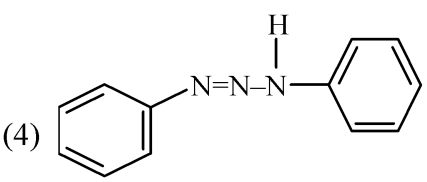
51. $C_6H_5NO_2 \xrightarrow{Sn + HCl} \text{"A"} \xrightarrow[H^{\oplus}]{C_6H_5\overset{\oplus}{N}_2\overset{\ominus}{Cl}} P$ (Yellow coloured compound) **[JEE-Mains-2021]**

Consider the above reaction, the Product "P" is :

(1) 

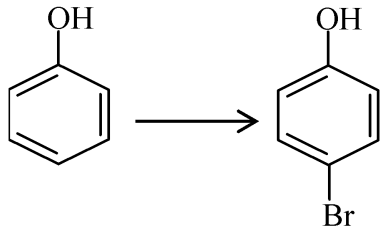
(2) 

(3) 

(4) 

AH0210

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\2-Aromatic\01_Th & Ex.p65

52.  **[JEE-Mains-2021]**

(Major product)

The given reaction can occur in the presence of :

(a) Bromine water (b) Br_2 in CS_2, 273 K

(c) $Br_2/FeBr_3$ (d) Br_2 in $CHCl_3$, 273 K

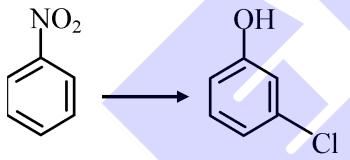
Choose the correct answer from the options given below :

(1) (b) and (d) only (2) (a) and (c) only

(3) (b), (c) and (d) only (4) (a), (b) and (d) only

AH0211

53. The correct sequence of correct reagents for the following transformation is :- **[JEE-Mains-2021]**



(1) (i) Fe, HCl (ii) Cl_2, HCl, (iii) $NaNO_2$, HCl, 0°C (iv) H_2O/H^+

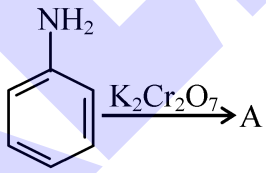
(2) (i) Fe, HCl (ii) $NaNO_2$, HCl, 0°C (iii) H_2O/H^+ (iv) Cl_2, $FeCl_3$

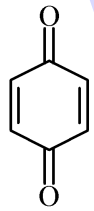
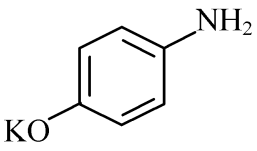
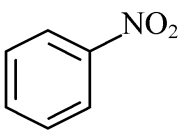
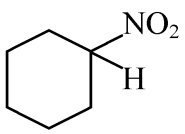
(3) (i) Cl_2, $FeCl_3$ (ii) Fe, HCl (iii) $NaNO_2$, HCl, 0°C (iv) H_2O/H^+

(4) (i) Cl_2, $FeCl_3$ (ii) $NaNO_2$, HCl, 0°C (iii) Fe, HCl (iv) H_2O/H^+

AH0212

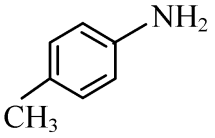
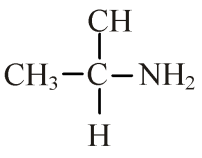
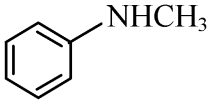
54. Identify A in the following reaction.

 **[JEE-Mains-2021]**

(1)  (2)  (3)  (4) 

AH0213

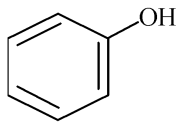
55. Which one of the following gives the most stable Diazonium salt ? **[JEE-Mains-2021]**

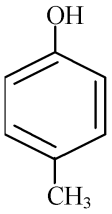
(1) CH_3-CH_2-CH_2-NH_2 (2)  (3)  (4) 

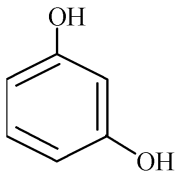
AH0214

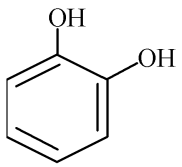
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\2-Aromatic\01_Th & Ex.p65

56. Which one of the following phenols does not give colour when condensed with phthalic anhydride in presence of conc.H_2SO_4 ? **[JEE-Mains-2021]**

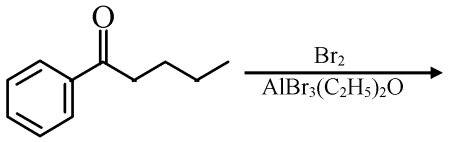
(1) 

(2) 

(3) 

(4) 

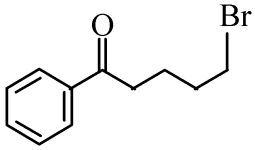
AH0215

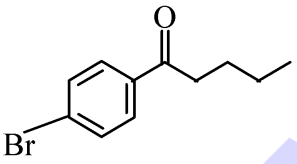
57. 

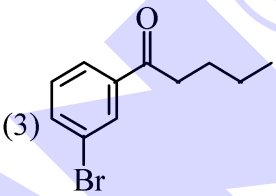
'A'
(Major product)

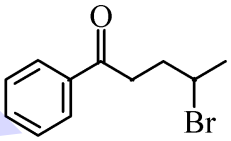
[JEE-Mains-2021]

Consider the given reaction, the product A is :

(1) 

(2) 

(3) 

(4) 

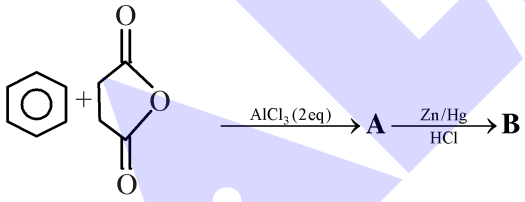
AH0216

58. The correct sequential addition of reagents in the preparation of 3-nitrobenzoic acid from benzene is: **[JEE-Mains-2021]**

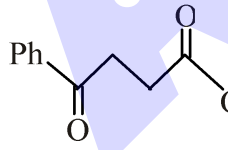
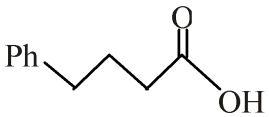
(1) $Br_2/AlBr_3$, HNO_3/H_2SO_4, Mg/ether, CO_2, H_3O^+
(2) $Br_2/AlBr_3$, NaCN, H_3O^+, HNO_3/H_2SO_4
(3) $Br_2/AlBr_3$, HNO_3/H_2SO_4, NaCN, H_3O^+
(4) HNO_3/H_2SO_4, $Br_2/AlBr_3$, Mg/ether, CO_2, H_3O^+

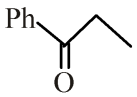
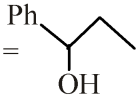
AH0218

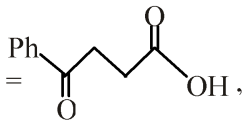
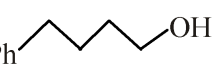
59. The structures of A and B formed in the following reaction are : [Ph = $-C_6H_5$]

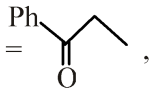
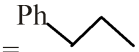


[JEE-Mains-2021]

(1) **A** =  , **B** = 

(2) **A** =  , **B** = 

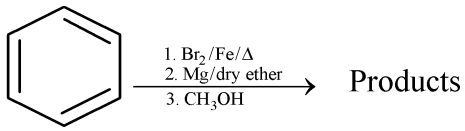
(3) **A** =  , **B** = 

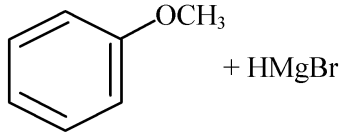
(4) **A** =  , **B** = 

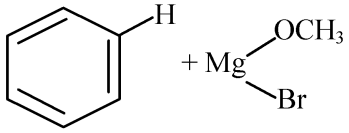
AH0219

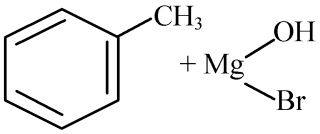
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\2-Aromatic\01_Th & Ex.p65

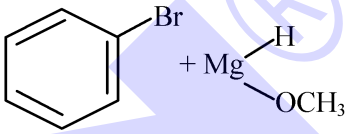
60. For the following : **[JEE-Mains-2021]**



(1) 

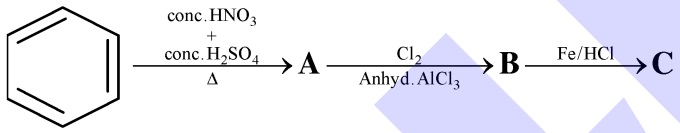
(2) 

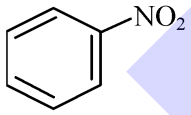
(3) 

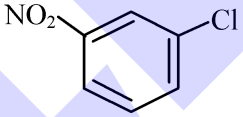
(4) 

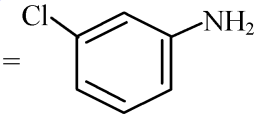
AH0220

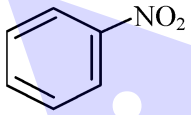
61. Identify correct A, B and C in the reaction sequence given below : **[JEE-Mains-2021]**

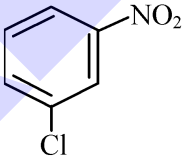


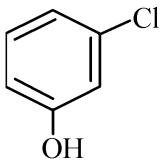
(1) A = 

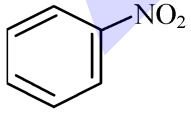
, B = 

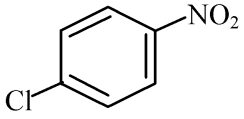
, C = 

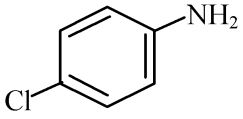
(2) A = 

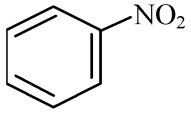
, B = 

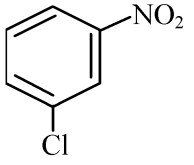
, C = 

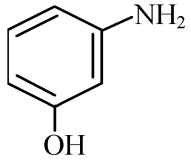
(3) A = 

, B = 

, C = 

(4) A = 

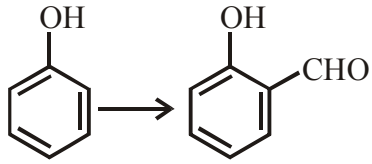
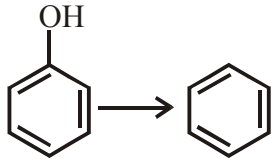
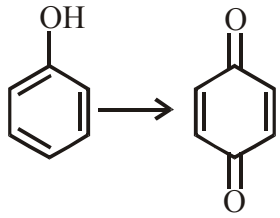
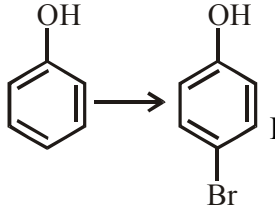
, B = 

, C = 

AH0221

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62. Match List I with List II **[JEE-Mains-2022]**

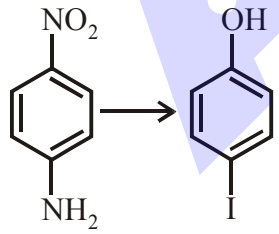
List I	**List II**
A. 	I. Br_2 in CS_2
B. 	II. $Na_2Cr_2O_7/H_2SO_4$
C. 	III. Zn
D. 	IV. $CHCl_3/NaOH$

Choose the correct answer from the options given below:

(1) A-IV, B-III, C-II, D-I
(2) A-IV, B-III, C-I, D-II
(3) A-II, B-III, C-I, D-IV
(4) A-IV, B-II, C-III, D-I

AH0240

63. The correct sequential order of the reagents for the given reaction is : **[JEE-Mains-2022]**

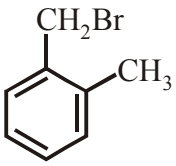


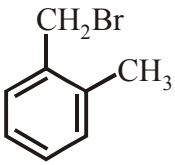
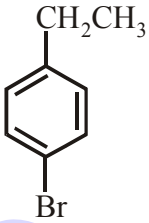
(1) HNO_2, Fe/H^+, HNO_2, KI, H_2O/H^+
(2) HNO_2, KI, Fe/H^+, HNO_2, H_2O/warm
(3) HNO_2, KI, HNO_2, Fe/H^+, H_2O/H^+
(4) HNO_2, Fe/H^+, KI, HNO_2, H_2O/warm

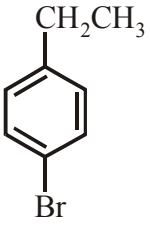
AH0241

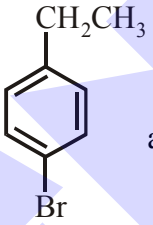
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep, Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex..p65

64. Two isomers (A) and (B) with Molar mass 184 g/mol and elemental composition C, 52.2%; H, 4.9% and Br 42.9% gave benzoic acid and p-bromobenzoic acid, respectively on oxidation with $KMnO_4$. Isomer 'A' is optically active and gives a pale yellow precipitate when warmed with alcoholic $AgNO_3$. Isomer 'A' and 'B' are, respectively : **[JEE-Mains-2022]**

(1) $H_3C\text{-}CHBr\text{-}C_6H_5$ and 

(2)  and 

(3) $H_3C\text{-}CHBr\text{-}C_6H_5$ and 

(4)  and $H_3C\text{-}CHBr\text{-}C_6H_5$

AH0242

65. Compound 'P' on nitration with dil. HNO_3 yields two isomers (A) and (B). These isomers can be separated by steam distillation. Isomers (A) and (B) show the intramolecular and intermolecular hydrogen bonding respectively. Compound (P) on reaction with conc. HNO_3 yields a yellow compound 'C', a strong acid. The number of oxygen atoms is present in compound 'C' _____.

[JEE-Mains-2022]

AH0243

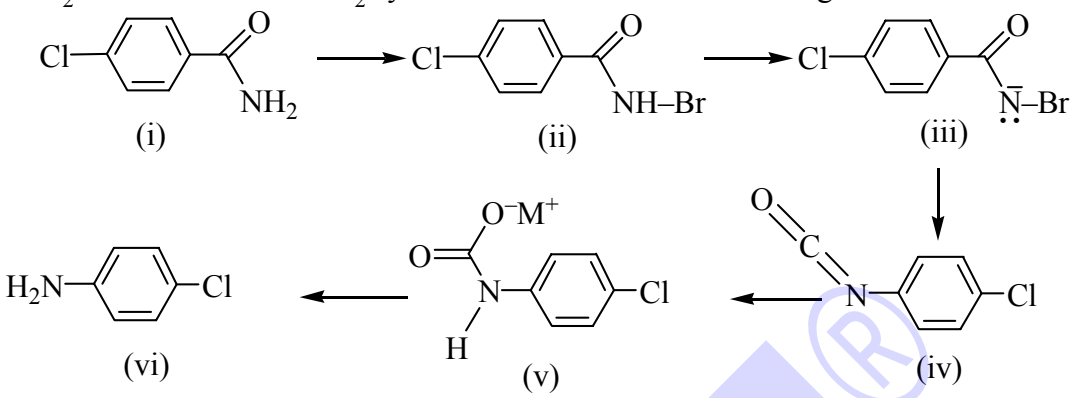
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\15-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

EXERCISE # (ADVANCE)

Question No. 1to 3 (3 questions)

Comprehension I

$RCONH_2$ is converted into RNH_2 by means of Hofmann bromamide degradation.



In this reaction, RCONHBr is formed from which this reaction has derived its name. Electron donating group at phenyl activates the reaction. Hofmann degradation reaction is an intramolecular reaction.

1. How can the conversion of (i) to (ii) be brought about? **[JEE 2006]**

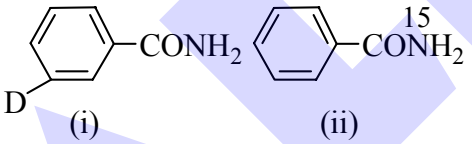
(A) KBr (B) KBr + CH_3ONa (C) KBr + KOH (D) Br_2 + KOH **AH0145**

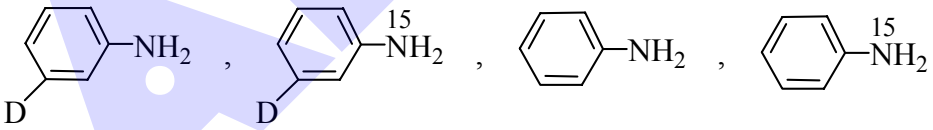
2. Which is the rate determining step in Hofmann bromamide degradation? **[JEE 2006]**

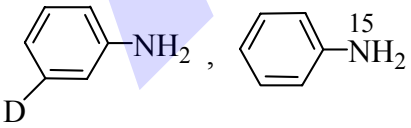
(A) Formation of (i) (B) Formation of (ii) (C) Formation of (iii) (D) Formation of (iv)

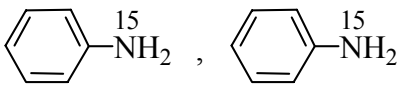
AH0145

3. What are the constituent amines formed when the mixture of (i) and (ii) undergoes Hofmann bromamide degradation ? **[JEE 2006]**



(A) 

(B) 

(C) 

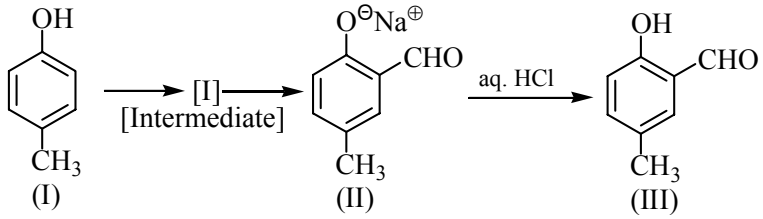
(D) 

AH0145

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Paragraph for Question Nos. 4 to 7 (4 questions)

Riemer-Tiemann reaction introduces an aldehyde group, on to the aromatic ring of phenol, ortho to the hydroxyl group. This reaction involves electrophilic aromatic substitution. This is a general method for the synthesis of substituted salicylaldehydes as depicted below.



4. Which one of the following reagents is used in the above reaction ? **[JEE 2007]**

(A) a NaOH + CH_3Cl (B) a NaOH + CH_2Cl_2

(C) a NaOH + $CHCl_3$ (D) aNaOH + CCl_4

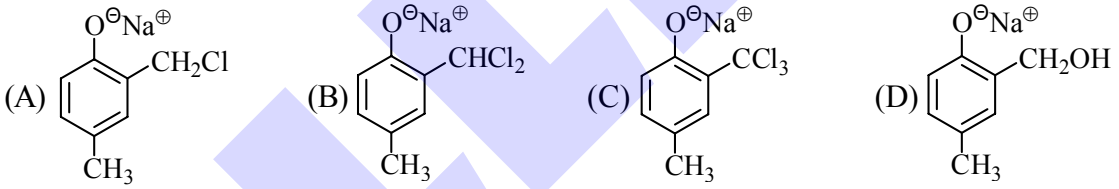
AH0146

5. The electrophile in this reaction is **[JEE 2007]**

(A) $:CHCl$ (B) $^{+}CHCl_2$ (C) $:CCl_2$ (D) $\bullet CCl_3$

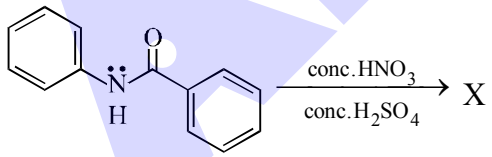
AH0146

6. The structure of the intermediate I is **[JEE 2007]**

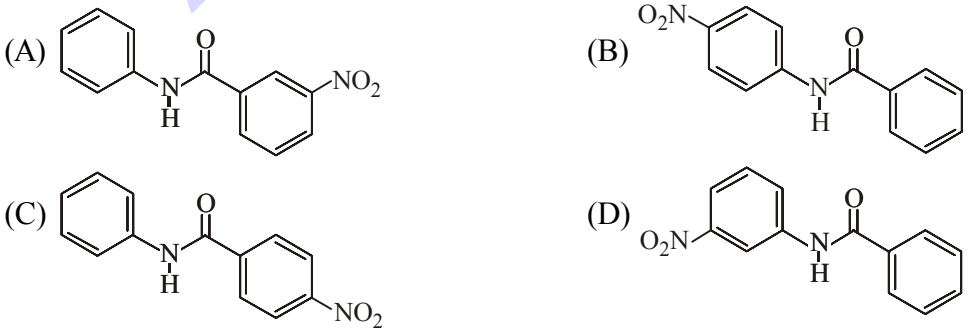


AH0146

7. In the following reaction, **[JEE 2007]**



the structure of the major product 'X' is



AH0147

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8. **Statement-1 :** Aniline on reaction with $NaNO_2$ / HCl at 0°C followed by coupling with β-naphthol gives a dark **blue** coloured precipitate. **[JEE 2008]**

and

Statement-2 : The colour of the compound formed in the reaction of aniline with $NaNO_2$ / HCl at 0° followed by coupling with β-naphthol is due to the extended conjugation.

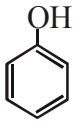
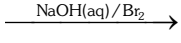
(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

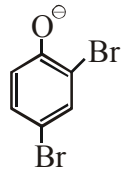
(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement1

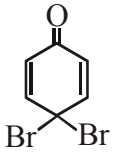
(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

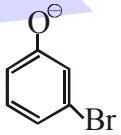
AH0149

9. In the reaction   the intermediate(s) is(are) – **[JEE 2009]**

(A) 

(B) 

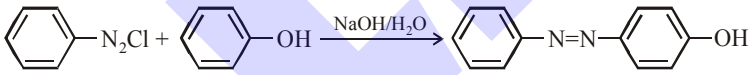
(C) 

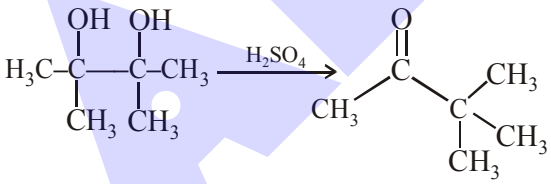
(D) 

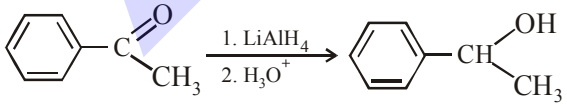
AH0150

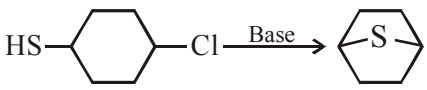
10. Match the reactions in Column–I with appropriate options in Column–II.

Column–I **Column–II** **[IIT-JEE-2010]**

(A) 

(B) 

(C) 

(D) 

(P) Racemic mixture

(Q) Addition reaction

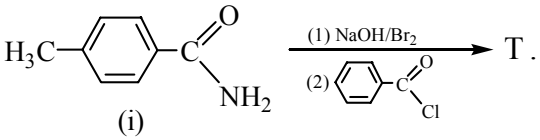
(R) Substitution reaction

(S) Coupling reaction

(T) Carbocation intermediate

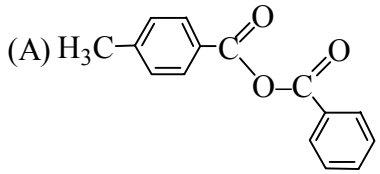
AH0153

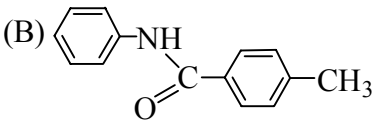
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th. & Ex..p65

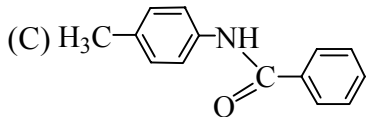
11. In the reaction 

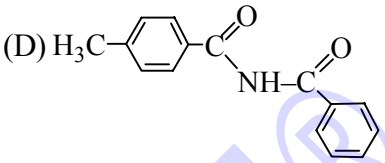
(i)

The structure of the Product T is : **[JEE 2010]**

(A) 

(B) 

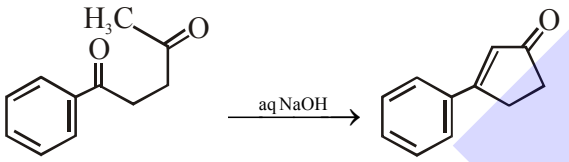
(C) 

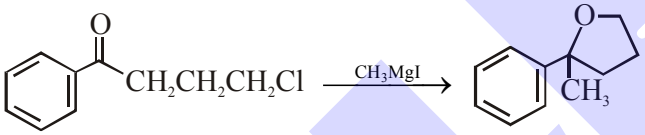
(D) 

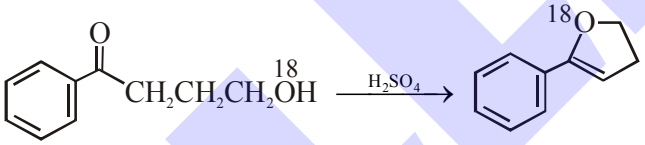
AH0154

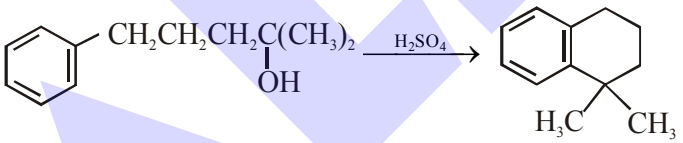
12. Match the reactions in **Column-I** with appropriate types of step/reactive intermediate involved in these reactions as given in **Column-II** **[IIT-JEE-2012]**

Column-I **Column-II**

(A) 

(B) 

(C) 

(D) 

(P) Nucleophilic substitution

(Q) Electrophilic substitution

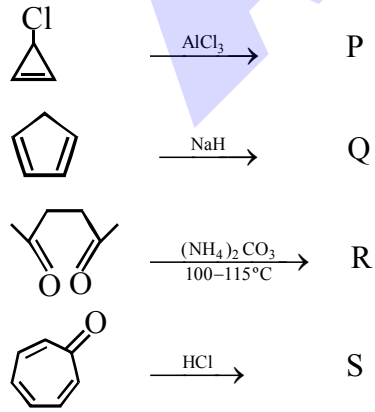
(R) Dehydration

(S) Nucleophilic addition

(T) Carbanion

AH0155

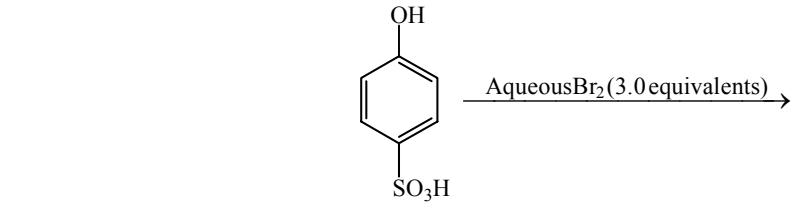
13. Among P, Q, R and S, the aromatic compound(s) is / are : **[JEE 2013]**

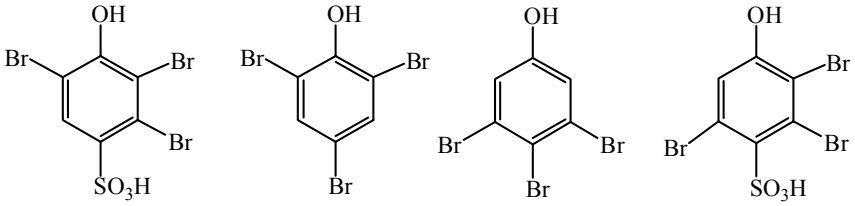


(A) P (B) Q (C) R (D) S **AH0156**

node06\B0BC-BD\Kota\JEE\Advanced\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

14. The major product(s) of the following reaction is (are) - **[JEE 2013]**



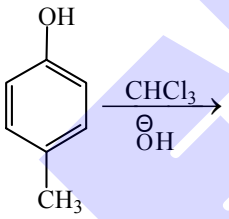


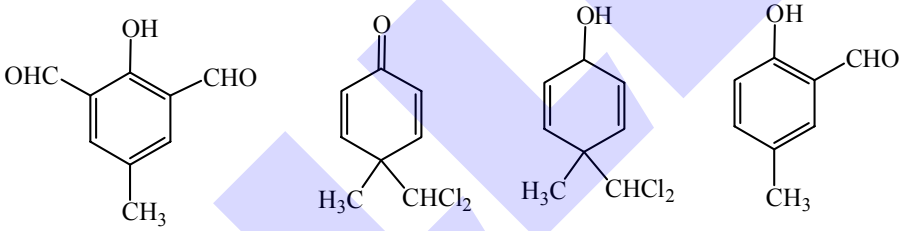
(P) (Q) (R) (S)

(A) P (B) Q (C) R (D) S

AH0157

15. In the following reaction, the product (s) formed is (are) **[JEE 2013]**





(P) (Q) (R) (S)

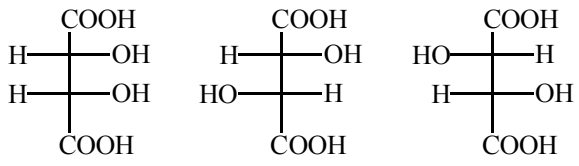
(A) P (major) (B) Q (minor) (C) R (minor) (D) S (major)

AH0158

Paragraph for Question 16 and 17

P & Q are isomers of dicarboxylic acid $C_4H_4O_4$. Both decolourize Br_2/H_2O, On heating P forms the cyclic anhydride.

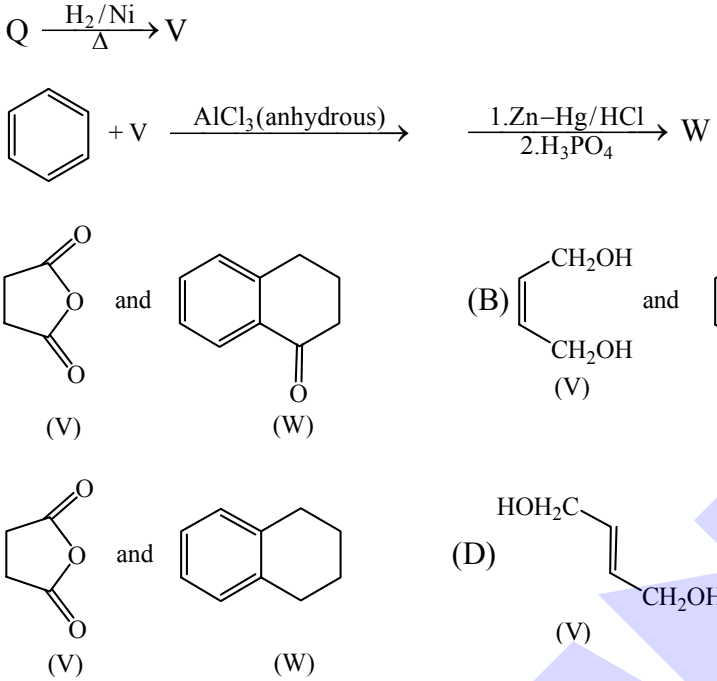
Upon treatment with dilute alkaline $KMnO_4$, P as well as Q could produce one or more than one from S, T and U. **[JEE 2013]**



(S) **(T)** **(U)**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex..p65

16. In the following reaction sequences V and W are , respectively -



AH0159

17. Compounds formed from P and Q are respectively

(A) Optically active S and optically active pair (T, U)

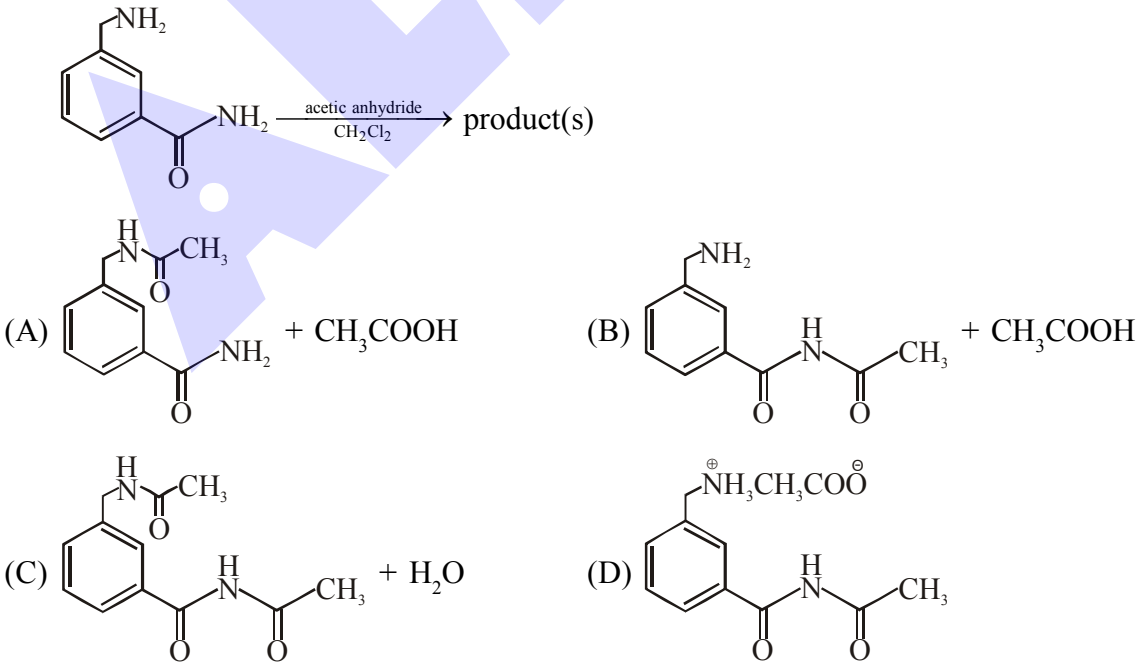
(B) Optically inactive S and optically inactive pair (T, U)

(C) Optically active pair (T, U) and optically active S

(D) Optically inactive pair (T, U) and optically inactive S

AH0159

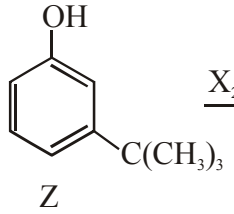
18. In the reaction shown below, the major product(s) formed is / are : **[JEE 2014]**



AH0160

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\15-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

19. The reactivity of compound Z with different halogens under appropriate conditions is given below : **[JEE 2014]**



Z

X_2 → mono halo substituted derivative when $X_2=I_2$

X_2 → di halo substituted derivative when $X_2=Br_2$

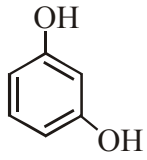
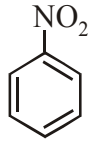
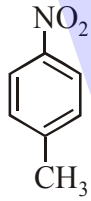
X_2 → tri halo substituted derivative when $X_2=Cl_2$

The observed pattern of electrop hilic substitution can be explained by -

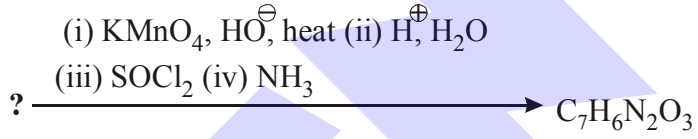
(A) The steric effect of the halogen
(B) The steric effect of the tert-butyl group
(C) The electronic effect of the phenolic group
(D) The electronic effect of the turt-butyl group

AH0161

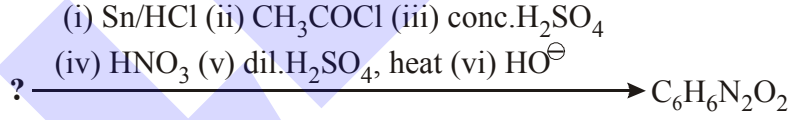
20. Match the four starting materials (P , Q, R, S) given in List I with the corresponding reaction scheme (I, II, III, IV) provided in List - II and select the correct answer using the code given below in lists. **[JEE 2014]**

List - I	**List - II**
(P) 	**(1) Scheme I**
(Q) 	**(2) Scheme II**
(R) 	**(3) Scheme III**
(S) 	**(4) Scheme IV**

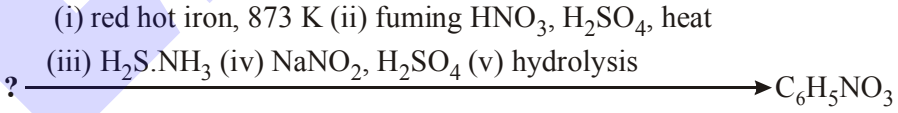
Scheme I



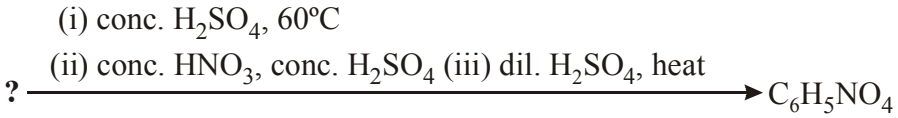
Scheme II



Scheme III



Scheme IV



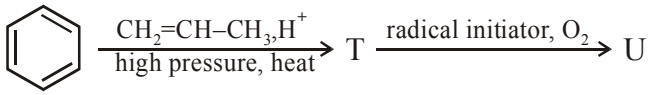
Code :

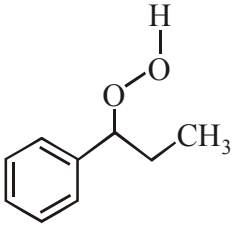
	P	Q	R	S
(A)	1	4	2	3
(B)	3	1	4	2
(C)	3	4	2	1
(D)	4	1	3	2

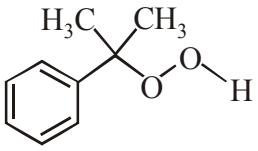
AH0162

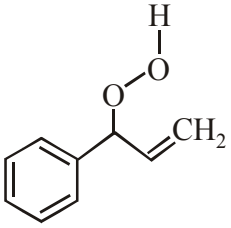
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th. & Ex..p65

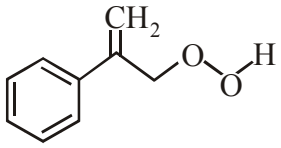
21. The major product U in the following reactions is : **[IIT 2015]**



(A) 

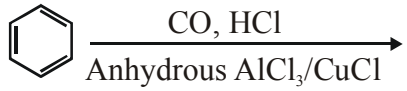
(B) 

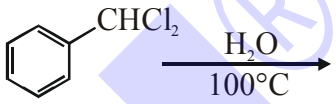
(C) 

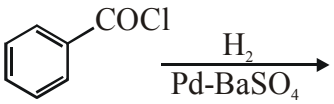
(D) 

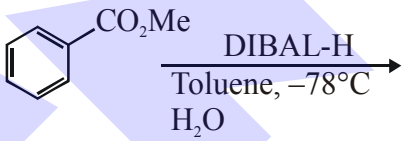
AH0163

22. Among the following the number of reaction(s) that produce(s) benzaldehyde is : **[IIT 2015]**

I. 

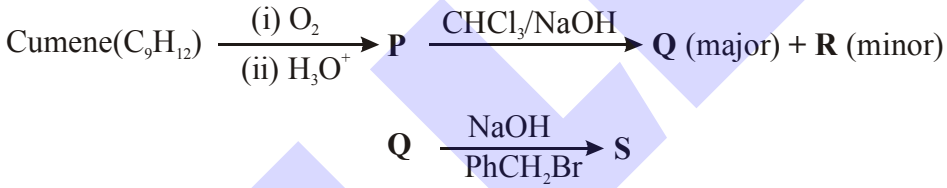
II. 

III. 

IV. 

AH0165

23. The correct statements(s) about of the following reaction sequence is(are) **[IIT 2016]**



(A) **R** is steam volatile

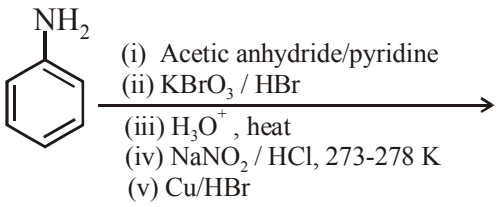
(B) **Q** gives dark violet coloration with 1% aqueous $FeCl_3$ solution

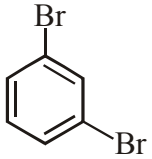
(C) **S** gives yellow precipitate with 2, 4,-dinitrophenylhydrazine

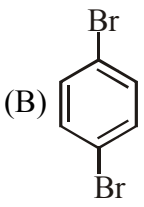
(D) **S** gives dark violet coloration with 1% aqueous $FeCl_3$ solution

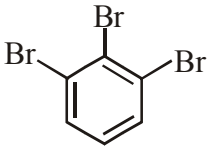
AH0166

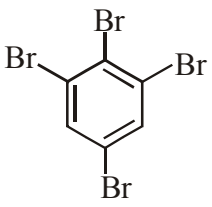
24. The product(s) of the following reaction sequence is(are) **[IIT 2016]**



(A) 

(B) 

(C) 

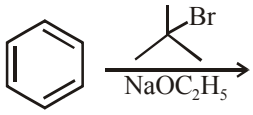
(D) 

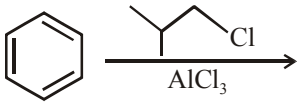
AH0167

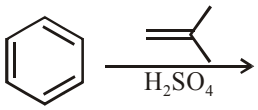
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Bimolek-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

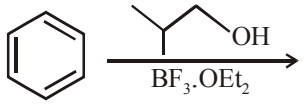
25. Among the following reaction(s) which gives (give) tert-butyl benzene as the major product is(are)

[IIT 2016]

(A) 

(B) 

(C) 

(D) 

AH0168

Answer 26, 27 and 28 by appropriately matching the information given in the three columns of the following table.

Columns 1, 2 and 3 contains starting materials, reaction conditions, , and type of reactions, respectively .

Column 1	Column 2	Column 3
(I) Toluene	(i) $NaOH/Br_2$	(P) Condensation
(II) Acetophenone	(ii) Br_2 / hν	(Q) Carboxylation
(III) Banzaldehyde	(iii)$(CH_3CO)_2O/CH_3COOK$	(R) Substitution
(IV) Phenol	(iv) $NaOH/CO_2$	(S) Haloform

26. For the synthesis of benzoic acid, the only CORRECT combination is

(A) (III) (iv) (R) (B) (IV) (ii) (P) (C) (I) (iv) (Q) (D) (II) (i) (S)

AH0171

27. The only CORRECT combination in which the reaction proceeds through radical mechanism is

(A) (I) (ii) (R) (B) (II) (iii) (R) (C) (III) (ii) (P) (D) (IV) (i) (Q)

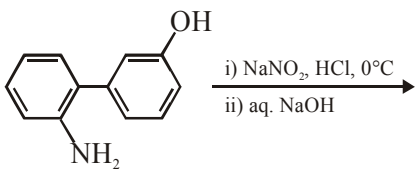
AH0171

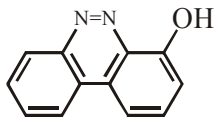
28. The only CORRECT combination that gives two different carboxylic acids is

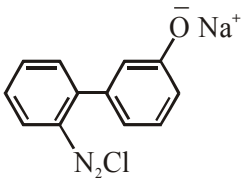
(A) (IV) (iii) (Q) (B) (III) (iii) (P) (C) (II) (iv) (R) (D) (I) (i) (S)

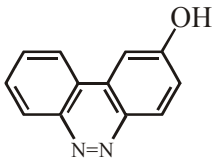
AH0171

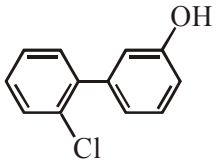
29. The major product of the following reaction is :



(A) 

(B) 

(C) 

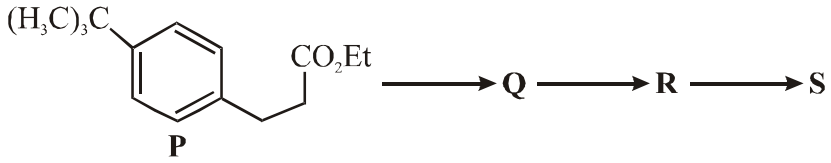
(D) 

AH0172

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th &Ex..p65

Paragraph for 30 & 32

The reaction of compound P with CH_3MgBr (excess) in $(C_2H_5)_2O$ followed by addition of H_2O gives **Q**, The compound **Q** on treatment with H_2SO_4 at 0ºC gives **R**. The reaction of **R** with CH_3COCl in the presence of anhydrous $AlCl_3$ in CH_2Cl_2 followed by treatment with H_2O produces compounds S. [Et it compounds **P** is ethyl group] **[IIT 2017]**



30. The reactions, **Q** to **R** and **R** to **S**, are -

(A) Dehydration and Friedel -Crafts acylation

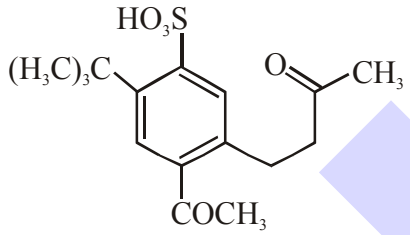
(B) Friedel-Crafts alkylation, dehydration and Friedel-Crafts acylation

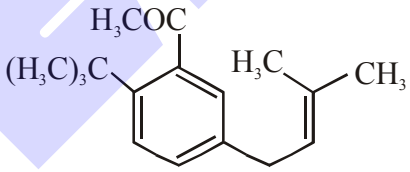
(C) Aromatic sulfonation and Friedel-Crafts acylation

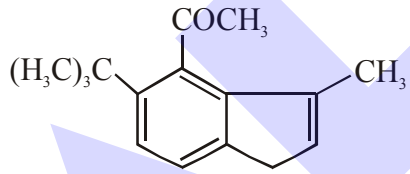
(D) Friedel-Crafts alkylation and Fridel-Crafts acylation

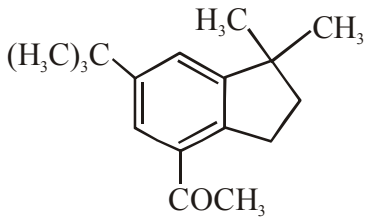
AH0173

31. The product **S** is -

(A) 

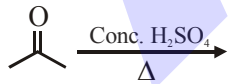
(B) 

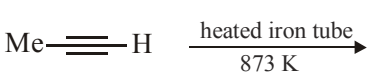
(C) 

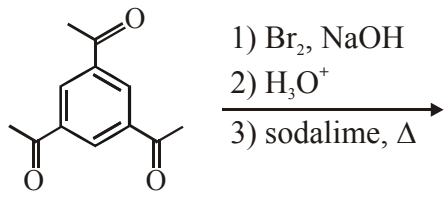
(D) 

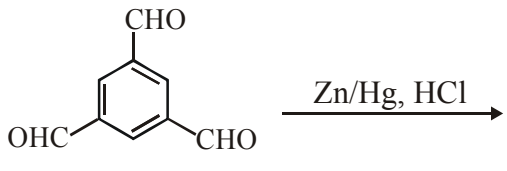
AH0173

32. The reaction(s) leading to the formation of 1,3,5-trimethylbenzene is (are) **[IIT 2018]**

(A) 

(B) 

(C) 

(D) 

AH0174

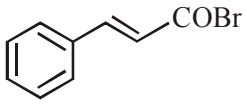
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\15-Carboxylic-Aromatic-Bimole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

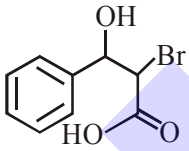
Paragraph "X"

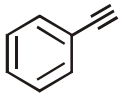
Treatment of benzene with CO/HCl in the presence of anhydrous $AlCl_3$/CuCl followed by reaction with Ac_2O/NaOAc gives compound X as the major product. Compound X upon reaction with Br_2/Na_2CO_3, followed by heating at 473 K with moist KOH furnishes Y as the major product. Reaction of X with H_2/Pd-C, followed by H_3PO_4 treatment gives Z as the major product. **[IIT 2018]**

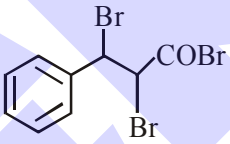
(There are two questions based on PARAGRAPH "X", the question given below is one of them)

33. The compound Y is :-

(A) 

(B) 

(C) 

(D) 

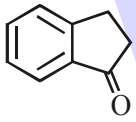
AH0175

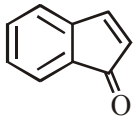
Paragraph "X"

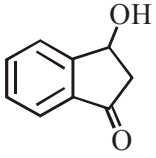
Treatment of benzene with CO/HCl in the presence of anhydrous $AlCl_3$/CuCl followed by reaction with Ac_2O/NaOAc gives compound X as the major product. Compound X upon reaction with Br_2/Na_2CO_3, followed by heating at 473 K with moist KOH furnishes Y as the major product. Reaction of X with H_2/Pd-C, followed by H_3PO_4 treatment gives Z as the major product.

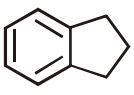
(There are two question based on PARAGARAPH "X", the question given below is one of them)

34. The compound Z is :-

(A) 

(B) 

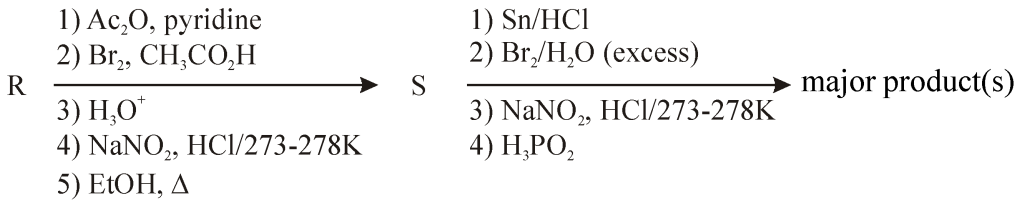
(C) 

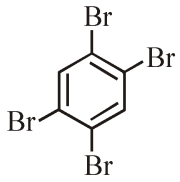
(D) 

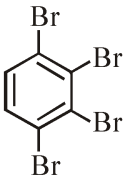
AH0175

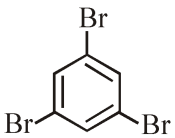
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th &Ex..p65

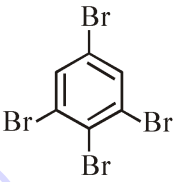
35. Aniline reacts with mixed acid (conc. HNO_3 and conc. H_2SO_4) at 288 K to give P (51%), Q (47%) and R (2%). The major product(s) the following reaction sequence is (are) :- **[IIT 2018]**



(A) 

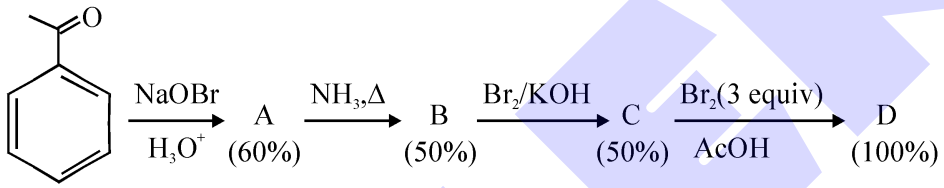
(B) 

(C) 

(D) 

AH0176

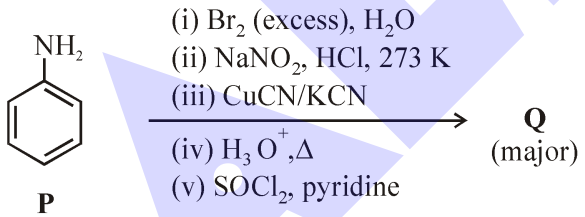
36. In the following reaction sequence, the amount of D (in g) formed from 10 moles of acetophenone is____. (Atomic weight in g mol^{-1}: H = 1, C = 12, N = 14, O = 16, Br = 80. The yield (%) corresponding to the product in each step is given in the parenthesis) **[IIT 2018]**



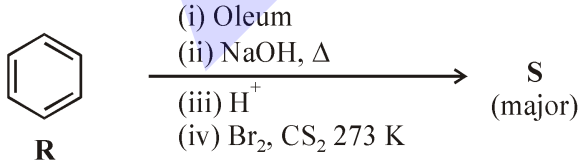
AH0177

37. Scheme 1 and 2 describe the conversion of **P** to **Q** and **R** to **S**, respectively. Scheme 3 describes the synthesis of **T** from **Q** and **S**. The total number of Br atoms in a molecule of **T** is _____

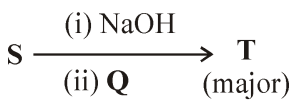
Scheme 1 : **[IIT 2019]**



Scheme 2 :



Scheme 3 :



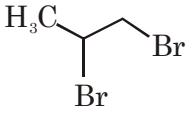
AH0178

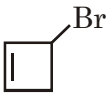
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\2-Aromatic\01_TH & Ex.p65

38. Choose the correct option(s) that give(s) an aromatic compound as the major product.

[IIT 2019]

(1)  + Cl_2(excess) $\xrightarrow{UV,\ 500K}$

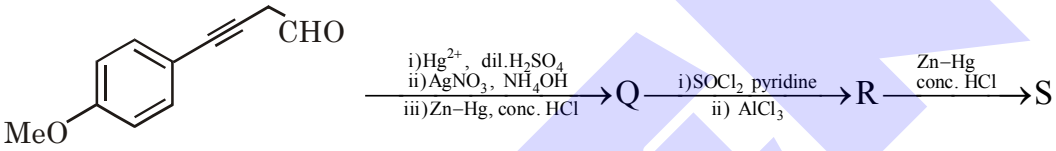
(2)  $\xrightarrow[iii)\ red\ hot\ iron\ tube,873\ K]{i)\ alc.KOH\ \ ii)\ NaNH_2}$

(3)  $\xrightarrow{NaOEt}$

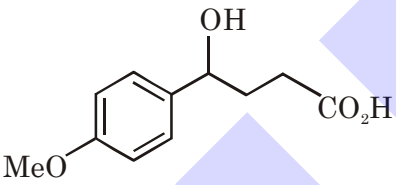
(4)  $\xrightarrow{NaOMe}$

AH0179

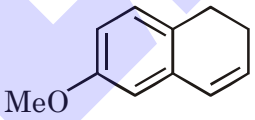
39. Choose the correct option(s) for the following reaction sequence **[IIT 2019]**



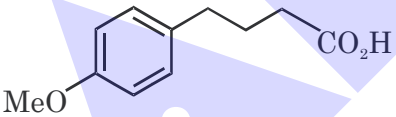
Consider Q, R and S as major products

(1) 

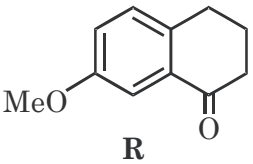
Q



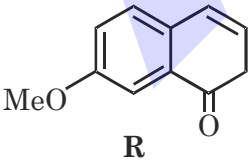
S

(2) 

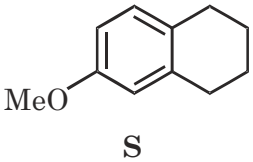
Q



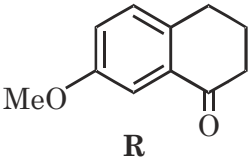
R

(3) 

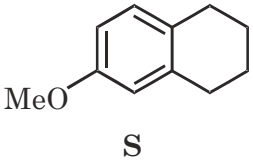
R



S

(4) 

R



S

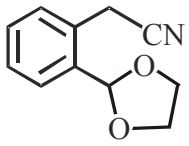
AH0180

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

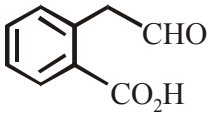
40. Answer the following by appropriately matching the lists based on the information given in the paragraph

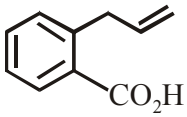
List-I includess starting materials and reagents of selected chemical reactions. List-II gives structures of compounds that may be formed as intermediate products and/or final products from the reactions of List-I **[IIT 2019]**

List-I **List-II**

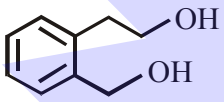
(I) 

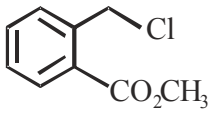
i) DIBAL-H
ii) dil, HCl
iii) $NaBH_4$
iv) conc. H_2SO_4

(P) 

(II) 

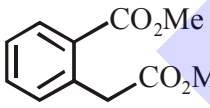
i) O_3
ii) Zn, H_2O
iii) $NaBH_4$
iv) conc. H_2SO_4

(Q) 

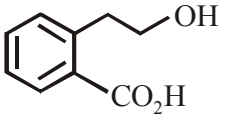
(III) 

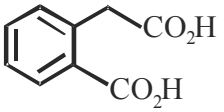
i) KCN
ii) H_3O^+, Δ
iii) $LiAlH_4$
iv) conc. H_2SO_4

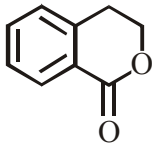
(R) 

(IV) 

i) $LiAlH_4$
ii) conc. H_2SO_4

(S) 

(T) 

(U) 

Which of the following options has correct combination considering List-I and List-II?

(1) (III), (S), (R)
(2) (IV), (Q), (R)
(3) (III), (T), (U)
(4) (IV), (Q), (U)

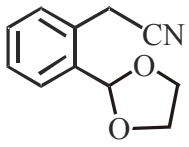
AH0181

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\15-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

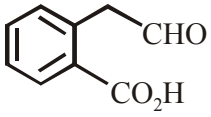
41. Answer the following by appropriately matching the lists based on the information given in the paragraph

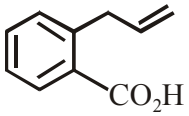
List-I includess starting materials and reagents of selected chemical reactions. List-II gives structures of compounds that may be formed as intermediate products and/or final products from the reactions of List-I **[IIT 2019]**

List-I **List-II**

(I) 

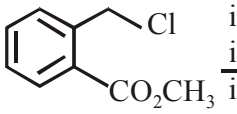
i) DlBAL-H
ii) dil, HCl
iii) $NaBH_4$
iv) conc. H_2SO_4

(P) 

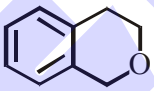
(II) 

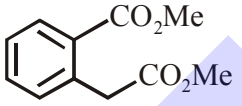
i) O_3
ii) Zn, H_2O
iii) $NaBH_4$
iv) conc. H_2SO_4

(Q) 

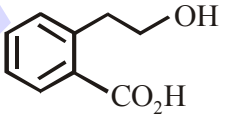
(III) 

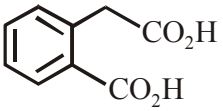
i) KCN
ii) H_3O^+, Δ
iii) $LiAlH_4$
iv) conc. H_2SO_4

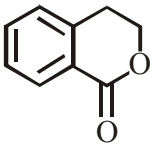
(R) 

(IV) 

i) $LiAlH_4$
ii) conc. H_2SO_4

(S) 

(T) 

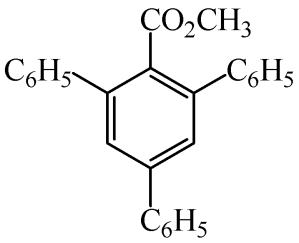
(U) 

Which of the following options has correct combination considering List-I and List-II?

(1) (I), (Q), (T), (U)
(2) (II), (P), (S), (U)
(3) (II), (P), (S), (T)
(4) (I), (S), (Q), (R)

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42. In the following reaction, compound **Q** is obtained from compound **P** via an ionic intermediate



P

$\xrightarrow{\text{conc. } H_2SO_4}$ **Q** (A coloured compound) **[IIT 2020]**

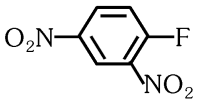
What is the degree of unsaturation of Q ? **AH0222**

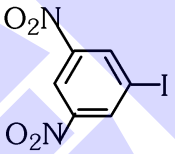
43. An organic compound ($C_8H_{10}O_2$) rotates plane-polarized light. It produces pink color with neutral $FeCl_3$ solution. What is the total number of all the possible isomers for this compound? **[IIT 2020]**

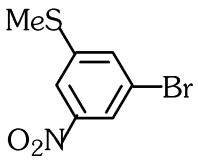
AH0223

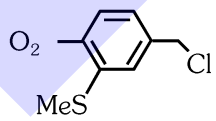
44. The reaction of Q with PhSNa yields an organic compound (major product) that gives positive Carius test on treatment with Na_2O_2 followed by addition of $BaCl_2$. The correct option(s) for Q is (are).

[IIT 2021]

(A) 

(B) 

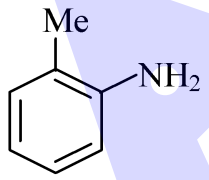
(C) 

(D) 

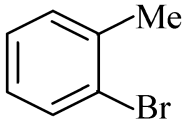
AH0224

45. The reaction sequence(s) that would lead to o-xylene as the major product is (are)

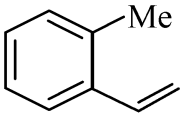
[IIT 2021]

(A) 

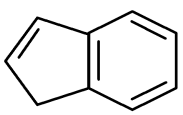
$\xrightarrow[\text{273 K}]{\text{1. } NaNO_2/HCl}$
$\xrightarrow{\text{2. CuCN}}$
$\xrightarrow{\text{3. DIBAL-H}}$
$\xrightarrow[\text{then } H_3O^+]{}$
$\xrightarrow{\text{4. } N_2H_4, \text{ KOH}}$
$\xrightarrow{\text{heat}}$

(B) 

$\xrightarrow{\text{1. Mg, } CO_2, H_3O^+}$
$\xrightarrow{\text{2. } SOCl_2}$
$\xrightarrow{\text{3. } H_2, \text{ Pd-BaSO}_4}$
$\xrightarrow{\text{4. Zn-Hg, HCl}}$

(C) 

$\xrightarrow[\text{ii. } H_2O_2, \text{ NaOH}]{\text{1. i. } BH_3}$
$\xrightarrow{\text{2. } PBr_3}$
$\xrightarrow{\text{3. Zn, dil. HCl}}$

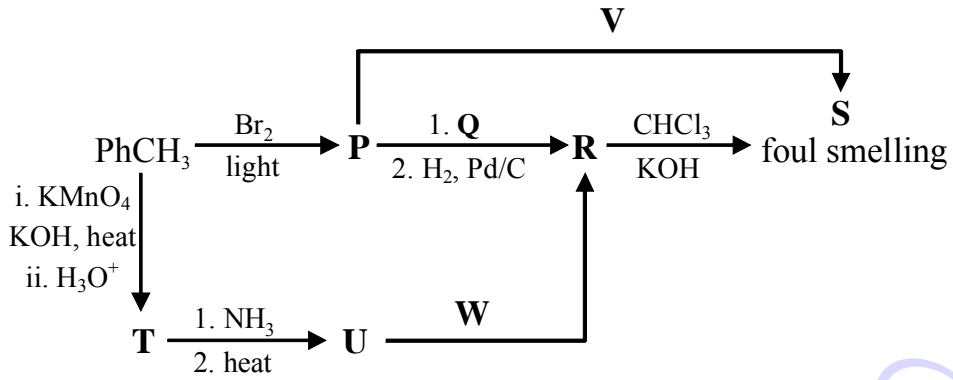
(D) 

$\xrightarrow{\text{1. } O_3, Zn/H_2O}$
$\xrightarrow{\text{2. } N_2H_4, \text{ KOH, heat}}$

AH0225

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46. Correct option(s) for the following sequence of reactions is(are) **[IIT 2021]**



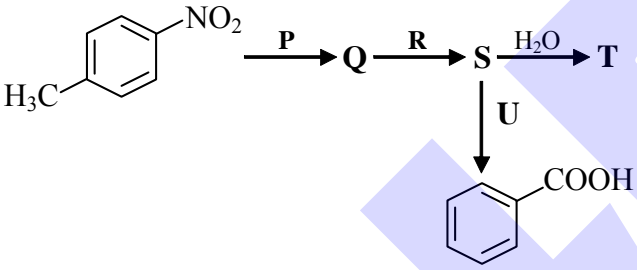
(foul smelling)

(A) Q = KNO_2, W = $LiAlH_4$ (B) R = benzenamine, V = KCN

(C) Q = $AgNO_2$, R = phenylmethanamine (D) W = $LiAlH_4$, V = AgCN

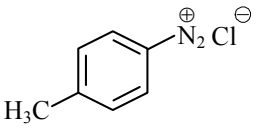
AH0226

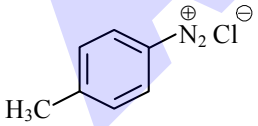
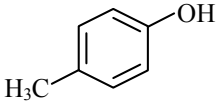
47. Consider the following reaction sequence, **[IIT 2022]**

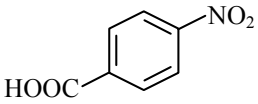
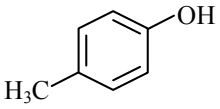


the correct option(s) is(are)

(A) P = H_2/Pd, ethanol R = $NaNO_2$/HCl U = 1. H_3PO_2
2. $KMnO_4$ - KOH, heat

(B) P = Sn/HCl R = HNO_2 S = 

(C) S =  T =  U = 1. CH_3CH_2OH
2. $KMnO_4$ - KOH, heat

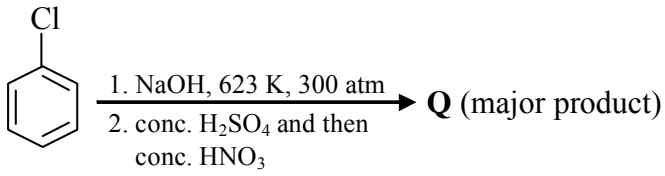
(D) Q =  R = H_2/Pd, ethanol T = 

AH0244

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48. The weight percentage of hydrogen in Q, formed in the following reaction sequence, is _____.

[IIT 2022]

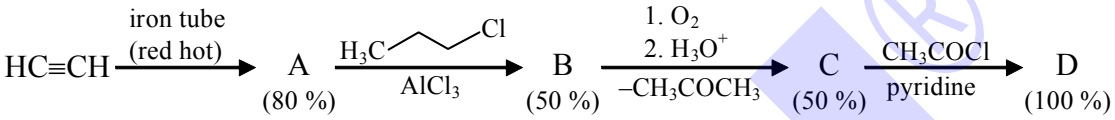


[Given : Atomic mass of H = 1, C = 12, N = 14, O = 16, S = 32, Cl = 35]

AH0245

49. If the reaction sequence given below is carried out with 15 moles of acetylene, the amount of the product D formed (in g) is _____.

[IIT 2022]



The yields of A, B, C and D are given in parentheses.
[Given : Atomic mass of H = 1, C = 12, O = 16, Cl = 35]

AH0246

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ANSWER-KEY

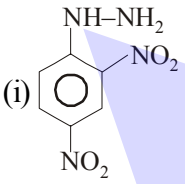
EXERCISE # O-I

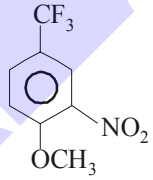
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	C	D	B	B	D	C	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	B	D	B	B	A	A	B	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	B	B	D	C	D	D	A	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	C	A	B	C	A	D	B	D	C

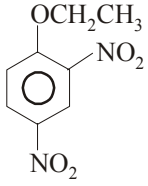
EXERCISE # O-II

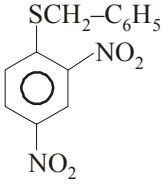
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	A,B,D	A,B,C	C,D	C,D	B,C	A,B	A,B,D	B,C,D	A,B,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,C,D	C	B	A,C,D	A,B,D	A	A	B,D	B,C,D	A,C,D
Que.	21	22	23	24				25		
Ans.	A,B,C,D	B,C	A,D	A→Q,R,S ; B →P,R,S ; C→R,S ; D→Q,R,S				A→R,T ; B →R,S ; C→Q,S ; D→P,R		
Que.	26			27						
Ans.	A → P,Q,R,S; B → P,S ; C → P,S ; D → Q,R,T			A → P,S; B → P,T; C → P,Q ; D → P,R						

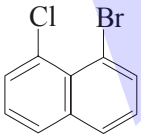
EXERCISE # S-I

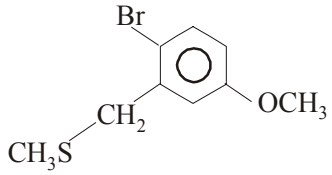
1. (i) 

(ii) 

(iii) 

(iv) 

2. (i) 

(ii) 

3. **(5)**
(i)–True, (ii)–True, (iii)–True, (iv)–True, (v)–True, (vi)–False, (vii)–False

4. **(3)**

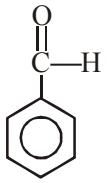
Sol. (i), (ii), (vi)

5. **(7)**

Sol. (a), (b), (c), (d), (e), (f), (h)

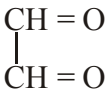
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex..pt5

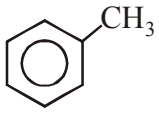
Solution for No. 6 & 7

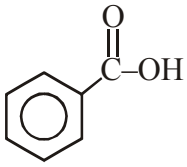
P = 

Q = $CH_3-\overset{O}{\overset{\|}{C}}-H$

R = Ph–CH = CH–CHO

S = 

T = 

U = 

V = 

W = HC ≡ CH

8. (B)

9. (A)

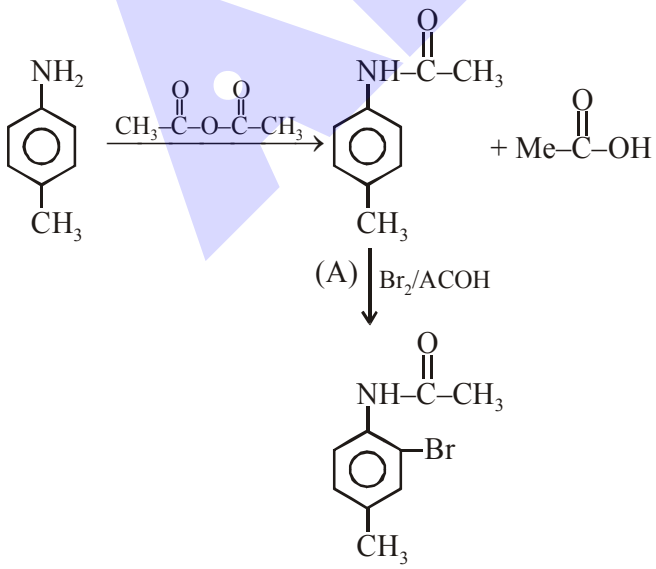
10. (150)

Sol. Ph — C(=O) — O — CH_2 — CH_3 (M.M = 150)

11. (3)

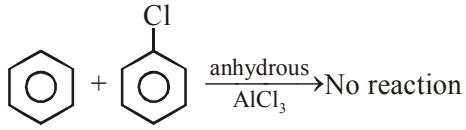
EXERCISE # (MAINS)

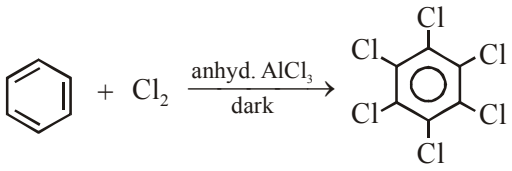
1. (3)	**2. (4)**	**3. (2)**	**4. (4)**	**5. (1)**	**6. (3)**
7. (4)	**8. (4)**	**9. (1)**	**10. (2)**	**11. (1)**	**12. (4)**
13. (2)	**14. (4)**	**15. (3)**	**16. (3)**	**17. (1)**	**18. (4)**
19. (1)	**20. (1)**	**21. (1)**	**22. (4)**		
23. (1)					

Sol. 

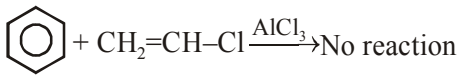
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\15-Carboxylic-Aromatic-Biamole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

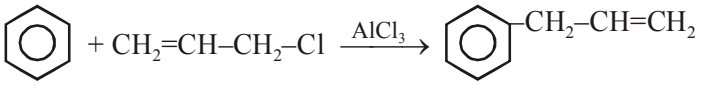
24. (2)

Sol. (a) 

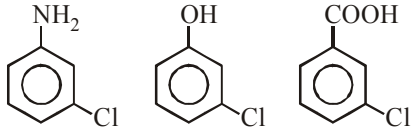
(b) 

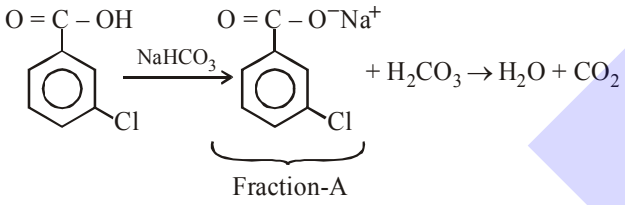
(electrophilic substitution)

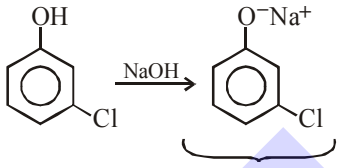
(c) 

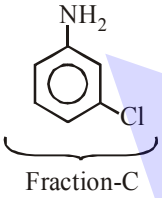
(d) 

25. (3)

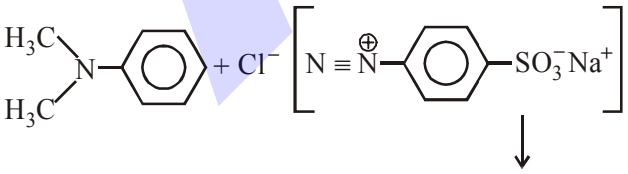
Sol. 

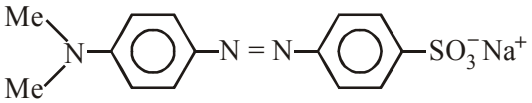






26. (1)



Sol. 

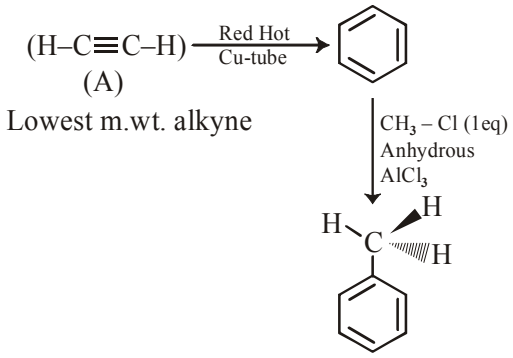
(Methyl orange)

It is an acid base indicator

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep, Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65

27 (13)

Sol.



Total 13 atom are present in same plane (7 carbon & 6 hydrogen atoms.)

28. (2) **29. (3)** **30. (2)** **31. (4)**

32. (1) **33. (3)** **34. (3)** **35. (2)**

36. (4) **37. (3)** **38. (2)** **39. (4)**

40. (3) **41. (4)** **42. (4)** **43. (2)**

44. (3) **45. (1)** **46. (1)** **47. (4)**

48. (4) **49. (1)** **50. (3)** **51. (2)**

52. (3) **53. (3)** **54. (1)** **55. (2)**

56. (2) **57. (3)** **58. (4)** **59. (1)**

60. (2) **61. (1)** **62. (1)** **63. (2)**

64. (3) **65. (7)**

EXERCISE # (ADVANCE)

Que.	1	2	3	4	5	6	7	8	9	
Ans.	D	D	B	C	C	B	B	D	A,B,C	
Que.	10				11	12				
Ans.	A → R, S, T ; B → T ; C → P, Q ; D → R				C	A → R,S,T ; B → P, S ; C → R, S ; D → Q, R				
Que.	13	14	15	16	17	18	19		20	
Ans.	A,B,C,D	B	B,D	A	B	A	A,B,C		C	
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	4	B,C	B	B,C,D	D	A	B	C	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	A,B,D	C	A	D	495	4.00	2,4	2,4	2
Que.	41	42	43	44	45	46	47	48	49	
Ans.	2	18.00	6	A,D	A,B	C,D	A,B,C	1.31	136	

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Everyday\Eng\2-Aromatic\01_Th & Ex.p65